

Depth and Lefschetz theorems in étale cohomology

SGA 2, Exposé XIV (M. Raynaud)

English draft

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About this draft

This is an unofficial English translation of Exposé XIV, “Depth and Lefschetz theorems in étale cohomology”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ XIV

Depth and Lefschetz theorems in étale cohomology

by Mme M. Raynaud¹

In no. 1, we define a notion of “*étale depth*” which is the analogue in étale cohomology of the notion of depth studied in ??, in the cohomology of coherent sheaves. After a technical part, we prove in no. 4 some “Lefschetz theorems”, the central theorem being 4.2. Let X be a scheme, Y a closed subset of X , U the complementary open set $X - Y$ and F an abelian sheaf on X , for the étale topology; in a general way the aim of the Lefschetz theorems is to show that, if F satisfies certain local conditions on U , expressible in terms of étale depth at the points of U , then, under certain additional conditions of a global nature on U (for example U *affine*), the natural map of étale cohomology groups

$$H^i(X, F) \longrightarrow H^i(Y, F|_Y)$$

is an isomorphism for values $i < n$, where n is a certain integer which is made explicit. Taking for F a constant sheaf, one thus obtains conditions for $\pi_0(X)$ to be equal to $\pi_0(Y)$ and conditions for the abelianized fundamental groups of X and Y to be the same. In no. 5, the introduction of a notion of “geometric depth” allows one to give useful special cases of the Lefschetz theorems (5.7). Finally in no. 6, we mention some conjectures, concerning in particular “noncommutative” variants of the theorems obtained.

1. Cohomological and homotopical depth

1.0. Let us fix the following notation. Let X be a scheme², Y a closed subset of X , U the complementary open set and $i : Y = X - U \rightarrow X$ the canonical immersion. Let $\underline{\Gamma}_Y$ be the functor which, to an abelian sheaf on X , associates the “sheaf of sections with support in Y ”, that is to say $\underline{\Gamma}_Y = i_* i^!$ (cf. SGA 4 IV 3.8 and VIII 6.6) and Γ_Y the functor $\Gamma \cdot \underline{\Gamma}_Y$ (where Γ is the functor “global sections”). Consider the derived category $D^+(X)$ and the derived functor $R\underline{\Gamma}_Y$ (resp. $R\Gamma_Y$) of $\underline{\Gamma}_Y$ (resp. of Γ_Y) (cf. [3]). Given a complex of abelian sheaves F on X , with degrees bounded below, one may regard it as an element of $D^+(X)$; we shall then denote by $\underline{H}_Y^p(F)$ the p -th cohomology sheaf of $R\underline{\Gamma}_Y(F)$ and by $H_Y^p(X, F)$ the p -th cohomology group of $R\Gamma_Y(F)$. The results of (SGA 4 V 4.3 and 4.4) extend trivially to $\underline{H}_Y^p(F)$ and $H_Y^p(X, F)$.

PROPOSITION 1.1. *Let X be a scheme, Y a closed subset of X , U the complementary open set and $i : U \rightarrow X$ the canonical immersion. Let F denote either a sheaf of sets on X , or a sheaf of groups on X , or a complex of abelian sheaves on X , with degrees bounded below. Let us fix the following notation: if $X' \rightarrow X$ is a morphism, U' and F' denote the inverse images of U and F on X' ; moreover, if $\bar{y}$ is a geometric point of X , $\bar{X}$ denotes the strict localization of X at $\bar{y}$ and $\bar{U}$ and $\bar{F}$ the inverse images of U and F in $\bar{X}$.*

1°) *Let F be a sheaf of sets on X and n an integer ≤ 2 ; then the following conditions are equivalent:*

¹After unpublished notes of A. Grothendieck.

²In accordance with the new terminology (cf. the re-edition of EGA I), we shall here call “scheme” what was previously called “prescheme” and “separated scheme” what was previously called “scheme”.

(i) *The canonical morphism*

$$F \longrightarrow i_* i^* F$$

is injective if $n \geq 1$, bijective if $n \geq 2$.

(ii) *For every scheme X' étale over X , the canonical morphism*

$$H^0(X', F') \longrightarrow H^0(U', F')$$

is injective if $n \geq 1$, bijective if $n \geq 2$.

Suppose moreover that the open set U is retrocompact in X ; then the preceding conditions are equivalent to the following:

(iii) *For every geometric point $\bar{y}$ of Y , the canonical morphism*

$$H^0(\bar{X}, \bar{F}) \longrightarrow H^0(\bar{U}, \bar{F})$$

is injective if $n \geq 1$, and bijective if $n \geq 2$.

2°) *Let F be a sheaf of groups on X and n an integer ≤ 3 ; then the following conditions are equivalent:*

(i) *The canonical morphism*

$$F \longrightarrow i_* i^* F$$

is injective if $n \geq 1$, bijective if $n \geq 2$, and if $n \geq 3$, in addition to the preceding conditions, the sheaf of pointed sets $R^1 i_(i^* F)$ is zero.*

(ii) *For every scheme X' étale over X , the canonical morphism*

$$H^0(X', F') \longrightarrow H^0(U', F')$$

is injective if $n \geq 1$, bijective if $n \geq 2$, and moreover the canonical morphism

$$H^1(X', F') \longrightarrow H^1(U', F')$$

is injective if $n \geq 2$, bijective if $n \geq 3$.

(ii bis) *Identical to (ii), except in the case $n = 2$ where one assumes only $H^0(X', F') \rightarrow H^0(U', F')$ bijective.*

Suppose moreover that U is retrocompact in X ; then the preceding conditions are also equivalent to the following:

(iii) *For every geometric point $\bar{y}$ of Y , the canonical morphism*

$$H^0(\bar{X}, \bar{F}) \longrightarrow H^0(\bar{U}, \bar{F})$$

is injective if $n \geq 1$, bijective if $n \geq 2$; finally if $n \geq 3$, in addition to the preceding conditions, $H^1(\bar{U}, \bar{F})$ is zero.

3°) *Let F be a complex of abelian sheaves, with degrees bounded below and n an integer; then the following conditions are equivalent:*

(i) *One has $H_Y^p(F) = 0$ for $p < n$ (cf. 1.0).*

(ii) *For every scheme X' étale over X , the canonical morphism*

$$H^p(X', F') \longrightarrow H^p(U', F')$$

is bijective for $p < n - 1$, injective for $p = n - 1$.

Suppose U retrocompact in X , then the preceding conditions are also equivalent to the following:

(iii) *For every geometric point $\bar{y}$ of Y , the canonical morphism*

$$H^p(\bar{X}, \bar{F}) \longrightarrow H^p(\bar{U}, \bar{F})$$

is bijective for $p < n - 1$, injective for $p = n - 1$.

In the case where F is an abelian sheaf and $n \geq 2$, the conditions (i) and (ii) are also equivalent to the following:

(ii bis) *For every scheme X' étale over X , the canonical morphism*

$$H^p(X', F') \longrightarrow H^p(U', F')$$

is bijective for $p < n - 1$.

PROOF. 1°) It is clear that (i) $\Leftrightarrow$ (ii). Let us show that, if U is retrocompact in X , (i) $\Leftrightarrow$ (iii). Indeed, (i) amounts to saying that, for every geometric point $\bar{y}$ of X , the morphism $F_{\bar{y}} \rightarrow (i_* i^* F)_{\bar{y}}$ is injective if $n \leq 1$ and bijective if $n \leq 2$ (SGA 4 VIII 3.6). As this morphism is in any case bijective when $\bar{y}$ is a geometric point of U , one may restrict to the geometric points $\bar{y}$ of Y . Now it follows from the fact that i is quasi-compact and from (SGA 4 VIII 5.3) that the morphism

$$F_{\bar{y}} \longrightarrow (i_* i^* F)_{\bar{y}}$$

is canonically identified with the morphism

$$H^0(\bar{X}, \bar{F}) \longrightarrow H^0(\bar{U}, \bar{F}),$$

whence the equivalence of (i) and (iii).

2°) (i) $\Rightarrow$ (ii). The assertions on H^0 follow from 1°. Let then i' be the canonical immersion of U' into X' ; the assertions on H^1 follow from the exact sequence (SGA 4 XII 3.2)

$$0 \longrightarrow H^1(X', i'_* i'^* F') \longrightarrow H^1(U', F') \longrightarrow H^0(X', R^1 i'_* (i'^* F')).$$

(ii bis) $\Rightarrow$ (i). By 1°, it suffices to show that, for $n \geq 3$, one has $R^1 i_*(i^* F) = 0$. Now $R^1 i_*(i^* F)$ is the sheaf associated to the presheaf $X' \mapsto H^1(U', F')$, that is to say, by hypothesis, the sheaf associated to the presheaf $X' \mapsto H^1(X', F')$, which is zero.

(i) $\Leftrightarrow$ (iii). Taking account of 1°, the only thing that remains to be seen is that the relation $R^1 i_*(i^* F) = 0$ is equivalent to the fact that $H^1(\bar{U}, \bar{F}) = 0$ for every geometric point $\bar{y}$ of Y . As $R^1 i_*(i^* F)$ is zero outside Y , it amounts to the same to say that $R^1 i_*(i^* F) = 0$ or that $(R^1 i_*(i^* F))_{\bar{y}} = 0$ for every geometric point $\bar{y}$ of Y . It then suffices to note that, i being quasi-compact, one has $(R^1 i_*(i^* F))_{\bar{y}} = H^1(\bar{U}, \bar{F})$ (SGA 4 VIII 5.3).

3°) (i) $\Rightarrow$ (ii). Let X' be a scheme étale over X ; one has the exact sequence (SGA 4 V 4.5)

$$(*) \quad \longrightarrow H_{Y'}^p(X', F') \longrightarrow H^p(X', F') \longrightarrow H^p(U', F') \longrightarrow;$$

hence (ii) is equivalent to $H_{Y'}^p(X', F') = 0$ for $p < n$ and for every scheme X' étale over X . Consider then the spectral sequence

$$E_2^{pq} = H^p(X', \underline{H}_Y^q(F)) \Longrightarrow H_{Y'}^p(X', F');$$

by hypothesis, $\underline{H}_Y^q(F) = 0$ for $q < n$, whence $E_2^{pq} = 0$ for $p + q < n$ and consequently $H_{Y'}^p(X', F') = 0$ for $p < n$.

(ii) $\Rightarrow$ (i). The sheaf $\underline{H}_Y^p(F)$ is associated to the presheaf $X' \mapsto H_{Y'}^p(X', F')$; as one has already remarked that (ii) is equivalent to the relation $H_{Y'}^p(X', F') = 0$ for $p < n$ and for every scheme X' étale over X , one indeed has $\underline{H}_Y^p(F) = 0$ for $p < n$.

(i) $\Leftrightarrow$ (iii). The sheaves $\underline{H}_Y^p(F)$ are concentrated on Y ; consequently it amounts to the same to say that $\underline{H}_Y^p(F) = 0$ or to say that, for every geometric point $\bar{y}$ of Y , the stalk $(\underline{H}_Y^p(F))_{\bar{y}}$ is zero. Now, i being quasi-compact, one deduces from (SGA 4 VIII 5.2) that one has $(\underline{H}_Y^p(F))_{\bar{y}} = \underline{H}_{\bar{Y}}^p(\bar{X}, \bar{F})$. The equivalence of (i) and (iii) follows from this, taking account of the analogue on $\bar{X}$ of the exact sequence (*).

(ii bis) $\Rightarrow$ (ii) in the case where F is an abelian sheaf. The only thing that remains to be shown is that $\underline{H}_Y^{n-1}(F) = 0$. Now, for $n > 2$, the sheaf $\underline{H}_Y^{n-1}(F)$ is associated to the presheaf $X' \mapsto H^{n-2}(U', F') = H^{n-2}(X', F')$ hence is indeed zero. The case $n = 2$ follows from the fact that $\underline{H}_Y^1(F)$ is the cokernel of the morphism $F \rightarrow i_* i^* F$. $\square$

DEFINITION 1.2. The notation is that of 1.1. One says that F is of étale Y -depth $\geq n$ and one writes

$$\text{prof}_Y(F) \geq n$$

if F satisfies the equivalent conditions (i) and (ii) of 1.1. If F is a complex of abelian sheaves, one calls the étale Y -depth of F the least upper bound of the n for which $\text{prof}_Y(F) \geq n$; one will use the same notation if F is a sheaf of sets resp. of not necessarily commutative groups (so that one then has $0 \leq \text{prof}_Y(F) \leq 2$, resp. $0 \leq \text{prof}_Y(F) \leq 3$, when the context does not allow confusion as to which of the three variants envisaged here one is using).

If $\mathbf{L}$ is a set of prime numbers, one says that the étale Y -depth for $\mathbf{L}$ of X is $\geq n$ and one writes

$$\mathrm{prof}_Y^{\mathbf{L}}(X) \geq n$$

if, for every constant sheaf of the form $\mathbf{Z}/\ell\mathbf{Z}$ with $\ell \in \mathbf{L}$, one has $\mathrm{prof}_Y(\mathbf{Z}/\ell\mathbf{Z}) \geq n$. One defines in the evident way the étale Y -depth for $\mathbf{L}$ of X . If $\mathbf{L} = \mathbf{P}$, the set of all prime numbers, and if there is no risk of confusion with the notation of (EGA IV 5.7.1) (relative to the case where $F = \mathcal{O}_X$), one omits $\mathbf{L}$ from the notation; otherwise one writes $\mathrm{prof\,ét}(X)$.

Finally one says that X is of homotopical Y -depth ≥ 3 for $\mathbf{L}$ and one writes

$$\mathrm{prof\,hop}_Y^{\mathbf{L}}(X) \geq 3$$

if, for every finite constant sheaf of $\mathbf{L}$ -groups F on X , one has $\mathrm{prof}_Y(F) \geq 3$. If $\mathbf{L} = \mathbf{P}$, one omits $\mathbf{L}$ from the notation.

COROLLARY 1.3. *Under the conditions of 1.1, if $\mathrm{prof}_Y(F) \geq n$, then, for every closed subset Z of Y , one has*

$$\mathrm{prof}_Z(F) \geq n.$$

Let us for example make the argument in the case where F is a complex of abelian sheaves with degrees bounded below. One uses 1.1 3°) (ii). Let $V = X - Z$ and consider, for every integer p , the sequence of morphisms

$$\mathrm{H}^p(X, F) \xrightarrow{f} \mathrm{H}^p(V, F) \xrightarrow{g} \mathrm{H}^p(U, F).$$

By hypothesis g and $f \circ g$ are bijective for $p < n - 1$ and injective for $p = n - 1$; the same is therefore true of f . As the argument is valid when one replaces X by a scheme X' étale over X , this proves 1.3.

COROLLARY 1.4. *The notation is that of 1.1 2°). If X' is a scheme over X , denote by Φ' the functor which associates to an étale covering of X' its restriction to U' , and $\Phi'_{F'}$, the functor which associates to a torsor³ under F' its restriction to U' . Then the following conditions are equivalent:*

- (i) *One has $\mathrm{prof}_Y(F) \geq 1$ (resp. $\mathrm{prof}_Y(F) \geq 2$, resp. $\mathrm{prof}_Y(F) \geq 3$).*
- (ii) *For every scheme X' étale over X , the functor $\Phi'_{F'}$ is faithful (resp. fully faithful, resp. an equivalence of categories).*

In particular, in order that one have $\mathrm{prof}_Y(X) \geq 1$ (resp. $\mathrm{prof}_Y(X) \geq 2$, resp. $\mathrm{prof\,hop}_Y(X) \geq 3$), it is necessary and sufficient that the functor Φ' be faithful (resp. fully faithful, resp. an equivalence of categories).

This indeed follows from 1.1 2°) (ii), taking account of the interpretation of $\mathrm{H}^1(X', F')$ as the set of classes (mod. isomorphisms) of torsors under F' (SGA 4 VII 2), and of the étale coverings Z of degree n of a scheme as associated to principal Galois coverings with group the symmetric group $\mathfrak{S}_n$, to Z being associated the covering $\underline{\mathrm{Isom}}_X(Z_0, Z)$, where Z_0 is the trivial covering of X of degree n .

COROLLARY 1.5. *Under the conditions of 1.1 3°), suppose that one has $\mathrm{prof}_Y(F) \geq n$; then one has*

$$\mathrm{H}_Y^n(X, F) \simeq \mathrm{H}^0(X, \underline{\mathrm{H}}_Y^n(F)).$$

The corollary follows from the spectral sequence

$$\mathrm{E}_2^{pq} = \mathrm{H}^p(X, \underline{\mathrm{H}}_Y^q(F)) \implies \mathrm{H}_Y^*(X, F).$$

Indeed one has by hypothesis $\mathrm{E}_2^{pq} = 0$ for $q < n$, whence it follows that

$$\mathrm{H}_Y^n(X, F) = \mathrm{E}_2^{0n} = \mathrm{H}^0(X, \underline{\mathrm{H}}_Y^n(F)).$$

³i.e. a “principal homogeneous bundle” in an older terminology.

REMARKS 1.6. a) The notion of Y -depth, in the form of the equivalent conditions (i) and (ii) of 1.1, makes sense for any site. In the particular case where X is a locally noetherian scheme, endowed with the Zariski topology, and F a sheaf of coherent $\mathcal{O}_X$ -modules, one finds the usual notion of Y -depth as the infimum of the depths at the points of Y (??).

b) For $n \leq 2$, the notion of étale Y -depth of X is independent of $\mathbf{L}$. For $n = 1$, it simply means that U is dense in X . Indeed this condition is necessary in order that one have $\text{prof}_Y(F) \geq 1$, and it is also sufficient because one may suppose X reduced, a case in which the condition U dense in X is preserved under étale change of base (EGA IV 11.10.5) (ii) b)). If U is retrocompact in X , the relation $\text{prof}_Y(X) \geq 1$ is also equivalent to saying Y contains no maximal point of X (EGA I 6.6.5). For $n = 2$ and U retrocompact in X , the condition $\text{prof}_Y(X) \geq 2$ is equivalent to the fact that, for every geometric point $\bar{y}$ of Y , $\bar{U}$ is connected and nonempty, that is to say that “ Y does not disconnect X , locally for the étale topology”.

c) If X is of Y -depth $\geq n$ for $\mathbf{L}$ and U retrocompact in X , then for every locally constant abelian sheaf of $\mathbf{L}$ -torsion F on X , one has $\text{prof}_Y(X) \geq n$. Indeed, the property $\text{prof}_Y(F) \geq n$ being local for the étale topology, one may suppose F constant; then F is a filtered inductive limit of sheaves that are finite sums of sheaves of the form $\mathbf{Z}/p^m\mathbf{Z}$, where m is an integer > 0 and $p \in \mathbf{L}$. Using 1.1 (iii) and (SGA 4 VII 3.3), one sees that one may reduce to the case where $F = \mathbf{Z}/p^m\mathbf{Z}$, then, by induction on m , to the case where $F = \mathbf{Z}/p\mathbf{Z}$ for which the assertion follows from the definition.

d) By 1.4, if one has $\text{prof}_Y(X) \geq 3$, the pair (X, Y) is *pure* in the sense of ?? ???. In fact the pure pairs that one encounters in practice (cf. ?? ??) satisfy the stronger condition of homotopical depth ≥ 3 , and this notion can therefore advantageously replace that of pure pair.

e) Let F be a complex of abelian sheaves and $T(F)$ the complex obtained by applying to F the translation functor ([3]); then one has evidently:

$$\text{prof}_Y(T(F)) = \text{prof}_Y(F) - 1.$$

f) Let us note that the recent work of Artin–Mazur ([1]) allows one to define the notion of homotopical depth $\geq n$, for any integer n (not only if $n \geq 3$).

g) Under the conditions of 1.1 3), in order that one have $\text{prof}_Y(F) = \infty$, it is necessary and sufficient that one have $F \xrightarrow{\sim} Ri_*(i^*F)$ in $D^+(X)$. Indeed, the $\underline{H}_Y^p(F)$ are the cohomology sheaves of the cone (= mapping cylinder) of the canonical morphism $F \rightarrow Ri_*i^*(F)$.

DEFINITION 1.7. Let X be a scheme, x a point of X , $\bar{x}$ a geometric point over x and $\bar{X}$ the strict localization of X at $\bar{x}$. As previously F denotes either a sheaf of sets on X , or a sheaf of groups on X , or a complex of abelian sheaves on X with degrees bounded below, $\bar{F}$ its inverse image on $\bar{X}$ and $\mathbf{L}$ a set of prime numbers. One says that F is of *étale depth* $\geq n$ at the point x (resp. that the *étale depth for $\mathbf{L}$ of X at x is* $\geq n$, resp. that the *homotopical depth for $\mathbf{L}$ of X at x is* ≥ 3) and one writes $\text{prof}_x(F) \geq n$ (resp. $\text{prof}_x^{\mathbf{L}}(X) \geq n$, resp. $\text{prof hop}_x^{\mathbf{L}}(X) \geq 3$) if one has $\text{prof}_{\bar{x}}(\bar{F}) > n$ (resp. $\text{prof}_{\bar{x}}^{\mathbf{L}}(\bar{X}) > n$, resp. $\text{prof hop}_{\bar{x}}^{\mathbf{L}}(\bar{X}) \geq 3$). One defines in the evident way the integer $\text{prof}_x^{\mathbf{L}}(X)$ and, if F is a complex of abelian sheaves, the integer $\text{prof}_x(F)$. If $\mathbf{L}$ is the set of all prime numbers, one omits $\mathbf{L}$ from the notation $\text{prof}_x^{\mathbf{L}}(X)$, unless there is a risk of confusion with the notation of (EGA IV 5.7.1), in which case one writes $\text{prof}_{\text{ét}_x}(X)$.

One then has the following *pointwise characterization* of depth:

THEOREM 1.8. *Let X be a scheme, Y a closed subset of X such that the open set $U = X - Y$ is retrocompact in X . If F is either a sheaf of sets on X , or a sheaf of groups on X , or a complex of abelian sheaves on X with degrees bounded below, then one has*

$$\text{prof}_Y(F) = \inf_{y \in Y} \text{prof}_y(F).$$

1.8.1. Let us first show that, for every point y of Y , one has the inequality $\text{prof}_y(F) \geq \text{prof}_Y(F)$. Let indeed $\bar{y}$ be a geometric point over y , $\bar{X}$ the strict localization of X at $\bar{y}$, $\bar{F}$

and $\overline{Y}$ the inverse images of F and Y on $\overline{X}$. One has by 1.7 and 1.3

$$\mathrm{prof}_y(F) = \mathrm{prof}_{\overline{y}}(\overline{F}) \geq \mathrm{prof}_{\overline{Y}}(\overline{F}) \geq \mathrm{prof}_Y(F),$$

the last inequality using the hypothesis “ U retrocompact”, via the conditions (iii) in 1.1 and the transitivity in the formation of strict localizations.

1.8.2. Conversely suppose that, for every point y of Y , one has $\mathrm{prof}_y(F) \geq n$ (n an integer) and let us show that one then has $\mathrm{prof}_Y(F) \geq n$.

Let us first recall the following well-known results (SGA 4 VIII):

LEMMA 1.8.2.1. *Let X be a scheme, F a sheaf of sets on X (resp. $G \rightarrow F$ a monomorphism of sheaves of sets on X). Then, in order that two sections s and s' of F coincide (resp. in order that a section s of F come from a section of G), it is necessary and sufficient that this be so locally. In particular, if s and s' are two sections of F , there exists a largest open set V of X on which they coincide (resp. if s is a section of F on X , there exists a largest open set V of X such that $s|_V$ comes from a section of G on V). This open set is also the set of points x of X such that, denoting by $\overline{x}$ a geometric point over x , the sections s and s' have the same image in the stalk $F_{\overline{x}}$ (resp. that the image of s in $F_{\overline{x}}$ comes from an element of $G_{\overline{x}}$).*

Let us return to the proof of 1.8.

1°) Case where F is a sheaf of sets. If $n = 1$, it suffices to show that the canonical morphism

$$H^0(X, F) \longrightarrow H^0(U, F)$$

is injective, the result still applying when one replaces X by a scheme étale over X . Let s and s' be two sections of F over X which have the same image in $H^0(U, F)$ and let V be the largest open set over which they are equal; one has evidently $V \supset U$. Suppose $V \neq X$ and let y be a maximal point of $X - V$, $\overline{y}$ a geometric point over y , $\overline{X}$ the strict localization of X at $\overline{y}$ and $\overline{V}$ and $\overline{F}$ the inverse images of V and F on $\overline{X}$. By the choice of y , one has $\overline{X} - \overline{y} = \overline{V}$ and by hypothesis, the morphism

$$H^0(\overline{X}, \overline{F}) \longrightarrow H^0(\overline{X} - \overline{y}, \overline{F}) = H^0(\overline{V}, \overline{F})$$

is injective. It follows that s and s' coincide at the point y , which is absurd. If $n = 2$, it suffices to show, taking account of the preceding, that the morphism

$$H^0(X, F) \longrightarrow H^0(U, F) = H^0(X, i_* i^* F)$$

is surjective (where i is the canonical immersion of U into X). Let s be a section of $i_* i^* F$ over X and V the largest open set over which it comes from a section of F . Suppose $V \neq X$ and let y be a maximal point of $X - V$; with the preceding notation, it follows from the hypothesis that the canonical morphism

$$H^0(\overline{X}, \overline{F}) \longrightarrow H^0(\overline{X} - \overline{y}, \overline{F}) = H^0(\overline{V}, \overline{F})$$

is bijective; consequently $s|_{\overline{V}}$ extends to the point $\overline{y}$, which is absurd and completes the proof in the case 1°).

2°) Case where F is a sheaf of groups. Taking account of 1°), the only thing that remains to be shown is that, in the case $n = 3$, the morphism

$$H^1(X, F) = H^1(X, i_* i^* F) \longrightarrow H^1(U, F)$$

is bijective. One already knows that it is injective thanks to 1°) and to 1.1 2°) (ii bis). For the surjectivity, one uses the exact sequence (SGA 4 XII 3.2)

$$0 \longrightarrow H^1(X, i_* i^* F) \longrightarrow H^1(U, F) \xrightarrow{d} H^0(X, R^1 i_*(i^* F)).$$

Let $s \in H^1(U, F)$ and $V \supset U$ the largest open set over which $d(s) = 0$; it is also the largest open set such that s comes from an element of $H^1(V, F)$. Suppose $V \neq X$ and let y be a

maximal point $X - V$; if $\bar{X}$ is the strict localization of X at a geometric point $\bar{y}$ over y , one has, with the obvious notation, the exact sequence

$$0 \longrightarrow H^1(\bar{X}, \bar{i}_*(\bar{i}^* \bar{F})) \longrightarrow H^1(\bar{U}, \bar{F}) \xrightarrow{\bar{d}} H^0(\bar{X}, R^1 \bar{i}_*(\bar{i}^* \bar{F})).$$

As $i : U \rightarrow X$ is a quasi-compact, $R^1 \bar{i}_*(\bar{i}^* \bar{F})$ is the inverse image of $R^1 i_*(i^* F)$ by the morphism $\bar{X} \rightarrow X$, whence $H^0(\bar{X}, R^1 \bar{i}_*(\bar{i}^* \bar{F})) = (R^1 i_*(i^* F))_{\bar{y}}$. By hypothesis and since $y \in Y$, the morphism

$$H^1(\bar{X}, \bar{F}) \longrightarrow H^1(\bar{V}, \bar{F})$$

is bijective. The image $\bar{s}$ of s in $H^1(\bar{U}, \bar{F})$, which extends to $\bar{V}$ by definition of V , therefore also extends to $\bar{X}$; it follows that $\bar{d}(\bar{s}) = 0$ hence the image of $d(s)$ in the geometric stalk $(R^1 i_*(i^* F))_{\bar{y}}$ is zero; but this contradicts the definition of V , whence the case 2°).

3°) Case where F is a complex of abelian sheaves, with degrees bounded below. One proceeds by induction on n . The conclusion holds for n sufficiently small, since F has degrees bounded below. Suppose therefore that $\text{prof}_Y(F) \geq n - 1$ and let us show that $\text{prof}_Y(F) \geq n$, knowing that, for every point y of Y , one has $\text{prof}_y(F) \geq n$. It suffices to see that the canonical morphism

$$(*) \quad H^{n-2}(X, F) \longrightarrow H^{n-2}(U, F)$$

is surjective and that

$$(**) \quad H^{n-1}(X, F) \longrightarrow H^{n-1}(U, F)$$

is injective (the result applying when one replaces X by a scheme étale over X).

a) Surjectivity of (*). The proof is analogous to that of 2°. Taking account of 1.5 and of (SGA 4 V 4.5), one has the exact sequence

$$H^{n-2}(X, F) \longrightarrow H^{n-2}(U, F) \xrightarrow{d} H_Y^{n-1}(X, F) = H^0(X, \underline{H}_Y^{n-1}(F)).$$

Let $s \in H^{n-2}(U, F)$ and $V \supset U$ the largest open set over which $d(s) = 0$, which is also the largest open set such that s extends to $H^{n-2}(V, F)$. Suppose $V \neq X$ and let y be a maximal point of $X - V$ and $\bar{X}$ the strict localization of X at a geometric point $\bar{y}$ over y . As $i : U \rightarrow X$ is quasi-compact, the formation of $\underline{H}_Y^{n-1}(F)$ commutes with the change of base $\bar{X} \rightarrow X$ and one therefore has (with the obvious notation) the exact sequence

$$H^{n-2}(\bar{X}, \bar{F}) \longrightarrow H^{n-2}(\bar{U}, \bar{F}) \xrightarrow{\bar{d}} H_{\bar{Y}}^{n-1}(\bar{X}, \bar{F}) = (\underline{H}_Y^{n-1}(F))_{\bar{y}},$$

the last equality resulting from the retrocompactness hypothesis on U .

Now one has by hypothesis the isomorphism

$$H^{n-2}(\bar{X}, \bar{F}) \xrightarrow{\sim} H^{n-2}(\bar{X} - \bar{y}, \bar{F}) = H^{n-2}(\bar{V}, \bar{F});$$

consequently the image $\bar{s}$ of s in $H^{n-2}(\bar{U}, \bar{F})$, which extends (by definition of V) to $H^{n-2}(\bar{V}, \bar{F})$, also extends to $H^{n-2}(\bar{X}, \bar{F})$; but this shows that $\bar{d}(\bar{s}) = 0$, that is to say that $d(s)$ is zero at $\bar{y}$, which is absurd.

b) Injectivity of (**). Using the surjectivity of (*), one obtains the exact sequence

$$0 \longrightarrow H^0(X, \underline{H}_Y^{n-1}(F)) \longrightarrow H^{n-1}(X, F) \longrightarrow H^{n-1}(U, F)$$

and one must show that every element $s \in H^0(X, \underline{H}_Y^{n-1}(F))$ is zero. Let V be the largest open set over which $s = 0$. Suppose that one has $V \neq X$ and let y be a maximal point of $X - V$, $\bar{X}$ a strict localization of X at a geometric point $\bar{y}$ over y . One has, by the induction hypothesis and by 1.8.1, the relation $\text{prof}_{\bar{Y}}(\bar{F}) \geq \text{prof}_Y(F) \geq n - 1$, whence the fact that the map e of the following diagram is injective:

$$H^0(\bar{X}, \underline{H}_Y^{n-1}(\bar{F})) = (\underline{H}_Y^{n-1}(F))_{\bar{y}} \xrightarrow{e} H^{n-1}(\bar{X}, \bar{F}) \xrightarrow{f} H^{n-1}(\bar{V}, \bar{F}).$$

The same holds for f by virtue of the hypothesis; the equality on the left results from the retrocompactness hypothesis on U . Let $\bar{s}$ be the image of s in $(\underline{H}_Y^{n-1}(F))_{\bar{y}}$; as s vanishes over

V , one has $f \cdot e(\bar{s}) = 0$, whence $\bar{s} = 0$, which contradicts the choice of y and completes the proof.

REMARK 1.9. A result analogous to 1.8 is doubtless valid in the case where one replaces the étale topos of a scheme X by a “topos locally of finite type”, that is to say one definable by a site locally of finite type (SGA 4 VI 1.1). To see this, one must use a result of P. Deligne (SGA 4 VI.9), asserting that there are “sufficiently many fiber functors” in such a topos.

We shall deduce from 1.8 some important cases in which one can determine the étale depth.

THEOREM 1.10. [*Cohomological semi-purity theorem*⁴] *Let X denote either a scheme smooth over a field k , or a regular excellent scheme (EGA IV 7.8.2) of characteristic zero (N.B. if one admits resolution of singularities in the sense of (SGA 4 XIX), it suffices to assume, more generally, that X is a regular excellent scheme of equal characteristic). Let Y be a closed subset of X and $\mathbf{L}$ the set of prime numbers distinct from the characteristic of X . Then one has*

$$\mathrm{prof}_{\mathbf{L}}^Y(X) = 2 \mathrm{codim}(Y, X).$$

PROOF. It follows from 1.8 that one has

$$\mathrm{prof}_{\mathbf{L}}^Y(X) = \inf_{y \in Y} \mathrm{prof}_y(X).$$

As on the other hand $\mathrm{codim}(Y, X) = \inf_{y \in Y} \dim \mathcal{O}_{X,y}$, one is reduced to showing that

$$\mathrm{prof}_{\mathbf{L}}^Y(X) = 2 \dim \mathcal{O}_{X,y},$$

which follows from (SGA 4 XVI 3.7 and XIX 3.2). $\square$

THEOREM 1.11 (Homotopical purity theorem⁵). *If X is a locally noetherian scheme which is regular (resp. whose local rings are complete intersections), Y a closed subset of X such that $\mathrm{codim}(Y, X) \geq 2$ (resp. $\mathrm{codim}(Y, X) \geq 3$), then one has*

$$\mathrm{prof} \mathrm{hop}_Y(X) \geq 3.$$

Indeed it follows from 1.8 that one has $\mathrm{prof} \mathrm{hop}_Y(X) = \inf_{y \in Y} \mathrm{prof} \mathrm{hop}_y(X)$. Now the strictly local rings of X at the different points of Y are regular rings of dimension ≥ 2 (resp. complete intersections and of dimension ≥ 3). It then follows from the purity theorem ?? ?? that $\mathrm{prof} \mathrm{hop}_y(X) \geq 3$, which proves the theorem.

⁴**N.D.E.** Gabber has since proved — in 1994 — Grothendieck’s absolute cohomological purity conjecture: if Y is a closed subscheme of absolutely noetherian schemes of pure codimension c and n an integer invertible on X , then $\mathrm{H}_Y^q(\Lambda)$ vanishes if $q \neq 2c$ and equals $\Lambda_Y(-c)$ (Tate twist) otherwise, where one has set $\Lambda = \mathbf{Z}/n\mathbf{Z}$. See (Fujiwara K., “A Proof of the Absolute Purity Conjecture (after Gabber)”, in *Algebraic geometry 2000, Azumino (Hotaka)*, Adv. Stud. in Pure Math., vol. 36, 2002, p. 153-183). For applications to the existence of the dualizing complex, see (SGA 5, Lect. Notes in Math., vol. 589, Springer-Verlag, 1977, p. 1672), exposé 1 and *loc. cit.*, §8. This conjecture had been proved in the case $n = \ell^\nu$ with ℓ a prime invertible on X sufficiently large, using K -theory in a crucial way (Thomason R.W., “Absolute cohomological purity”, *Bull. Soc. Math. France* **112** (1984), no. 3, p. 397–406). One finds K -theory again in Gabber’s proof via the Atiyah–Hirzebruch–Thomason spectral sequence relating étale cohomology and K -theory, a method already used in Thomason’s approach. Besides this result, the other fundamental argument is the generalization of the Lefschetz theorem cited in note (8), page 18.

⁵**N.D.E.** recently, de Jong and Oort have obtained the following purity statement: let $\tilde{S} \rightarrow S$ be a resolution of singularities of the spectrum S of a normal noetherian local ring of dimension 2 and let U be the complement of the closed point s in S . Suppose moreover that $k(s)$ is algebraically closed. Then, for every prime number p , in particular if S is of characteristic p , the restriction morphism $H_{\mathrm{ét}}^1(\tilde{S}, \mathbf{Q}_p) \rightarrow H_{\mathrm{ét}}^1(U, \mathbf{Q}_p)$ is bijective (de Jong A.J. & Oort F., “Purity of the stratification by Newton polygons”, *J. Amer. Math. Soc.* **13** (2000), no. 1, p. 209–241, theorem 3.2). If $k = \mathbf{C}$ and A is the completion of a surface singularity, this result is due to Mumford (see page ??, [?]).

EXAMPLE 1.12. Let X be a locally noetherian scheme, Y a closed subset of X and $n = 1$ or 2 . Then, if one has $\text{prof}_Y(\mathcal{O}_X) \geq n$ ($\text{prof}_Y(\mathcal{O}_X)$ denoting the Y -depth in the sense of coherent sheaves (cf. 1.6 a)), one also has $\text{prof}_Y(X) \geq n$; this is obvious for $n = 1$ and, for $n = 2$, it is nothing other than Hartshorne's theorem (?? ??). On the other hand the analogous assertion is false for $n \geq 3$. Take for example an affine space of dimension ≥ 3 over a field of characteristic $\neq 2$ and let the group $\mathbf{Z}/2\mathbf{Z}$ act by symmetry with respect to the origin. Let X be the quotient and $Y = \{x\}$ the image of the origin in X . Then $\mathcal{O}_{X,x}$ is a Cohen-Macaulay ring, hence one has $\text{prof}_x(\mathcal{O}_X) \geq 3$; but the affine space with the origin removed is an étale covering of $X - \{x\}$ which does not extend to an étale covering of X ; hence one has by 1.4 $\text{prof}_Y(X) = 2$.

The following theorem is the analogue of (EGA IV 6.3.1):

THEOREM 1.13. *Let $f : X \rightarrow S$ be a morphism of schemes, Y a closed subset of X , Z a closed subset of S , such that $f(Y) \subset Z$. One assumes that the local rings of X at the different points of Y are noetherian and that the open sets $X - Y$ and $S - Z$ are retrocompact in X and S respectively. Let p, q, r be integers such that $p \geq -r$, $q \geq 0$, $\mathbf{L}$ a set of prime numbers and F a complex of abelian sheaves of $\mathbf{L}$ -torsion on S , such that the cohomology sheaves $H^i(F)$ vanish for $i < -r$. One assumes that*

- a) *The morphism f is locally $(p + q + r - 2)$ -acyclic for $\mathbf{L}$ (SGA 4 XV 1.11).*
- b) *One has*

$$\text{prof}_Z(F) \geq p.$$

- c) *For every point s of Z , one has*

$$\text{prof}_{Y_s}^{\mathbf{L}}(X_s) \geq q.$$

Then one has

$$\text{prof}_Y(f^*F) \geq p + q.$$

We shall need the following lemma:

LEMMA 1.13.1. *Let $\mathbf{L}$ be a set of prime numbers, n and r integers, $f : X \rightarrow S$ a morphism locally n -acyclic for $\mathbf{L}$. Let F be a complex of abelian sheaves, with cohomology sheaves of $\mathbf{L}$ -torsion, such that $H^i(F) = 0$ for $i < -r$, Z a closed subset of S such that $S - Z$ is retrocompact in S and $T = f^{-1}(Z)$. Then the canonical morphism*

$$f^*(\underline{H}_Z^i(F)) \longrightarrow \underline{H}_T^i(f^*F)$$

is bijective for $i < n - r + 2$ and injective for $i = n - r + 2$.

Put $U = S - Z$ and $V = X - T$, so that one has the cartesian square

$$\begin{array}{ccc} V & \xrightarrow{g} & U \\ k \downarrow & & \downarrow j \\ X & \xrightarrow{f} & S \end{array}$$

Consider the following commutative diagram whose rows are exact

$$\begin{array}{ccccccc} \longrightarrow & f^*(\underline{H}_Z^i(F)) & \longrightarrow & f^*(\underline{H}^i(F)) & \longrightarrow & f^*(\underline{H}^i(\mathbf{R}j_*(j^*F))) & \longrightarrow \\ & \downarrow & & \downarrow \wr & & \downarrow & \\ \longrightarrow & \underline{H}_T^i(f^*F) & \longrightarrow & \underline{H}^i(f^*F) & \longrightarrow & \underline{H}^i(\mathbf{R}k_*(k^*f^*F)) & \longrightarrow ; \end{array}$$

it follows that one is reduced to showing that the morphism

$$f^*(\underline{H}^i(\mathbf{R}j_*(j^*F))) \longrightarrow \underline{H}^i(\mathbf{R}k_*(k^*f^*F))$$

is bijective for $i < n - r + 1$ and injective for $i = n - r + 1$. Now such a morphism comes from the following morphism between hypercohomology spectral sequences

$$\begin{array}{ccc} f^* E_2^{p,q} = f^*(R^p j_* (\underline{H}^q(j^* F))) & \Longrightarrow & f^*(\underline{H}^*(R j_*(j^* F))) \\ \downarrow & & \downarrow \\ E_2^{p,q} = R^p k_*(\underline{H}^q(k^* f^* F)) & \Longrightarrow & \underline{H}^*(R k_*(k^* f^* F)). \end{array}$$

As j is quasi-compact, it follows from (SGA 4 XV 1.10) that the morphism $f^*(E_2^{p,q}) \rightarrow E_2^{p,q}$ is bijective for $p \leq n$ and injective for $p = n + 1$; in particular it is bijective for $p + q \leq n - r$ and injective for $p + q = n - r + 1$. The conclusion follows at once.

Let us return to the proof of 1.13. Let $T = f^{-1}(Z)$. By 1.13.1 and condition a), the canonical morphism $f^*(\underline{H}_Z^i(F)) \rightarrow \underline{H}_T^i(f^* F)$ is an isomorphism for $i \leq p + q$. It therefore follows from b) that $\underline{H}_T^i(f^* F) = 0$ for $i < p$ and, for $i < p + q$, $\underline{H}_T^i(f^* F)$ restricted to T is the inverse image of a sheaf G^i on Z . Let

$$f_T : T \longrightarrow Z$$

be the restriction of f to T . It then follows from c) and from the corollary which follows that $\underline{H}_Y^j(f_T^*(G^i)) = 0$ for $j < q$. One concludes from this that

$$\underline{H}_Y^j(\underline{H}_T^i(f^* F)) = 0 \text{ for } i + j < p + q,$$

for the inequality $i + j < p + q$ implies either $i < p$ and then $\underline{H}_T^i(f^* F) = 0$, or $j < q$ and then $\underline{H}_Y^j(\underline{H}_T^i(f^* F)) = 0$. Given that one has, with the notations of 1.0, $\underline{\Gamma}_Y = \underline{\Gamma}_Y \cdot \underline{\Gamma}_T$, one has the spectral sequence

$$(1.13.2) \quad E_2^{i,j} = \underline{H}_Y^j(\underline{H}_T^i(f^* F)) \Longrightarrow \underline{H}_Y^*(f^* F);$$

as $E_2^{i,j} = 0$ for $i + j < p + q$, one sees that $\underline{H}_Y^k(f^* F) = 0$ for $k < p + q$. The theorem will thus be proved if one proves the following corollary (which is the special case of 1.13 obtained by making $Z = S$, $r = p = 0$ therein and F reduced to degree 0).

COROLLARY 1.14. *Let $f : X \rightarrow S$ be a morphism, Y a closed subset such that the complementary open set $X - Y$ is retrocompact in X and that the local rings of X at the different points of Y are noetherian. Let $\mathbf{L}$ be a set of prime numbers, q an integer and F an abelian sheaf of $\mathbf{L}$ -torsion on S . Suppose that f is locally $(q - 2)$ -acyclic for $\mathbf{L}$ and that, for every point s of S , one has $\text{prof}_{Y_s}^{\mathbf{L}}(X_s) \geq q$. Then one has $\text{prof}_Y(f^* F) \geq q$.*

1°) Reduction to the case where X and S are strictly local schemes, f a local morphism and Y reduced to a closed point of X .

By 1.8, to establish 1.14, one must show that one has for every point y of Y :

$$\text{prof}_y(f^* F) \geq q.$$

Let $s = f(y)$, $\bar{s}$ a geometric point over s , $\bar{y}$ a geometric point over y and over $\bar{s}$, $\bar{X}$ and $\bar{S}$ the strict localizations of X and S at $\bar{y}$ and $\bar{s}$ respectively, $\bar{f} : \bar{X} \rightarrow \bar{S}$ the canonical morphism and $\bar{F}$ the inverse image of F on $\bar{S}$. As one has the relation $\text{prof}_y(f^* F) = \text{prof}_{\bar{y}}(\bar{f}^* \bar{F})$, it suffices to show that the hypotheses of 1.14 are preserved when one replaces f (resp. Y , resp. F) by $\bar{f}$ (resp. $\{\bar{y}\}$, resp. $\bar{F}$). The retrocompactness condition follows from the noetherian hypothesis on $\mathcal{O}_{X,x}$, implying that $\bar{X}$ is noetherian. By (SGA 4 XV 1.10 (i)), $\bar{f}$ is still locally $(q - 2)$ -acyclic for $\mathbf{L}$. On the other hand the fibre $(\bar{X})_{\bar{s}}$ of $\bar{X}$ over $\bar{s}$ is identified with the strict localization of X_s at $\bar{y}$, hence satisfies the relation $\text{prof}_{\bar{y}}((\bar{X})_{\bar{s}}) \geq q$. As an analogous relation is trivially satisfied for the fibres of the $\bar{S}$ -scheme $\bar{X}$, other than the closed fibre, this completes the reduction.

2°) Case where X and S are strictly local, f a local homomorphism and Y reduced to the closed point of X . Let

$$g : Y = X - \{y\} \longrightarrow S$$

be the structure morphism of U . One must show that the canonical morphism

$$u_i : H^i(X, f^*F) \longrightarrow H^i(U, f^*F)$$

is bijective for $i \leq q - 2$ and injective for $i = q - 1$. Consider the commutative diagram

$$\begin{array}{ccc} H^i(X, f^*F) & \xrightarrow{u_i} & H^i(U, f^*F) \\ & \swarrow v_i \quad \searrow w_i & \\ & H^i(S, F) & \end{array}$$

The morphism v_i is obviously bijective for every i . On the other hand g is locally $(q - 2)$ -acyclic for $\mathbf{L}$; moreover its fibres are $(q - 2)$ -acyclic for $\mathbf{L}$, as follows from the fact that $\text{prof}_y(X_s) \geq q$ and that the fibres of f are $(q - 2)$ -acyclic for $\mathbf{L}$; as g is quasi-compact since X is noetherian, it follows from (SGA 4 XV 1.16) that g is $(q - 2)$ -acyclic for $\mathbf{L}$. Consequently w_i , hence also u_i , is bijective for $i \leq q - 2$ and injective for $i = q - 1$, which completes the proof of 1.14.

COROLLARY 1.15. *Let $f : X \rightarrow S$ be a morphism of schemes, $\mathbf{L}$ a set of prime numbers, m and r integers and F a complex of abelian sheaves of $\mathbf{L}$ -torsion on S such that $\underline{H}^i(F) = 0$ for $i < -r$. Let x be a point of X , $s = f(x)$ and suppose that the local ring $\mathcal{O}_{X,x}$ is noetherian. Then, if f is locally m -acyclic for $\mathbf{L}$, one has the relation*

$$(*) \quad \text{prof}_x(f^*F) \geq \inf(\text{prof}_s(F) + \text{prof}_x^{\mathbf{L}}(X_s), n) \text{ where } n = m - r + 2.$$

In particular, if $n \geq \text{prof}_s(F) + \text{prof}_x^{\mathbf{L}}(X_s)$, for example if f is locally acyclic for $\mathbf{L}$, one has

$$(**) \quad \text{prof}_x(f^*F) \geq \text{prof}_s(F) + \text{prof}_x^{\mathbf{L}}(X_s).$$

If $\mathbf{L}$ is reduced to a single element ℓ and if one has $n \geq \text{prof}_s(F) + \text{prof}_x^{\mathbf{L}}(X_s)$, the preceding inequality is an equality.

One reduces to the case where s and x are closed points, by taking the strict localizations of S and x at geometric points $\bar{s}$ over s and $\bar{x}$ over x and over $\bar{s}$. If one has the inequality $n \geq \text{prof}_s(F) + \text{prof}_x^{\mathbf{L}}(X_s)$, then $(*)$ is obtained from 1.13 by making $p = \text{prof}_s(F)$ and $q = \text{prof}_x^{\mathbf{L}}(X_s)$ therein (the hypothesis that $S - \{s\}$ is retrocompact in S follows from the fact that $X - X_s$ is retrocompact in X and that f is surjective since it is (-1) -acyclic (except perhaps if the conclusion of 1.15 is empty)). If one has $n < \text{prof}_s(F) + \text{prof}_x^{\mathbf{L}}(X_s)$, the inequality $(*)$ is still obtained from 1.13 by making for example $p = \text{prof}_s(F)$ and $q = n - p$ therein. It remains to prove the last assertion. Let $p = \text{prof}_s(F)$ and $q = \text{prof}_x(X_s)$; it follows from (1.13.2) that one has

$$\underline{H}_x^{p+q}(f^*(F)) \simeq \underline{H}_x^q(f^*(\underline{H}_s^p(F))).$$

As $\text{prof}_s(F) = p$, the sheaf $\underline{H}_s^p(F)$ is a sheaf of ℓ -torsion, constant on s , different from zero. Consequently the sheaf $G = f^*(\underline{H}_s^p(F))$ is a sheaf of ℓ -torsion, constant on X_s , nonzero, hence contains a subsheaf isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$; as $\underline{H}_x^q(\mathbf{Z}/\ell\mathbf{Z})$ is different from zero, one indeed has $\underline{H}_x^q(G) \neq 0$.

COROLLARY 1.16. *Let $f : X \rightarrow S$ be a regular morphism of excellent schemes (EGA IV 7.8.2) of characteristic zero, ℓ a prime number and F a complex of sheaves of ℓ -torsion on S . Let $x \in X$, $s = f(x)$; then one has*

$$\text{prof}_x(f^*F) = \text{prof}_s(F) + 2 \dim(\mathcal{O}_{X,x}).$$

Indeed f is locally acyclic (SGA 4 XIX 4.1). It then follows from 1.15 that one has

$$\text{prof}_x(f^*F) = \text{prof}_s(F) + \text{prof}_x(X_s).$$

Now one has by 1.10

$$\text{prof}_x(X_s) = 2 \dim \mathcal{O}_{X_s,x},$$

whence the result.

REMARK 1.17. It follows from 1.15 that 1.13 remains true when one replaces b) and c) by the conditions:

b') For every point $s \in f(Y)$, one has $\text{prof}_s(F) \geq p$.

c') For every point $x \in Y$, if $s = f(x)$, one has $\text{prof}_x^{\mathbf{L}}(X_s) \geq q$.

In the case of a sheaf of sets or of groups, one has the following theorem analogous to 1.13.

THEOREM 1.18. *Let $f : X \rightarrow S$ be a morphism of schemes, Y a closed subset of X such that $X - Y$ is retrocompact in X and that, for every point x of Y , the local ring $\mathcal{O}_{X,x}$ is noetherian.*

1°) *Let F be a sheaf of sets on S and n an integer equal to 1 or 2. Suppose that f is locally $(n - 2)$ -acyclic and that, for every point s of $f(Y)$, one has:*

$$\text{prof}_{Y_s}(X_s) + \text{prof}_s(F) \geq n.$$

Then one has:

$$\text{prof}_Y(f^*F) \geq n.$$

2°) *Let $\mathbf{L}$ be a set of prime numbers and F a sheaf of ind- $\mathbf{L}$ -groups. Suppose that f is locally 1-aspherical for $\mathbf{L}$ (SGA 4 XV 1.11) and that, for every point s of $f(Y)$, one has:*

$$\text{prof hop}_{Y_s}^{\mathbf{L}}(X_s) + \text{prof}_s(F) \geq 3.$$

Then, one has:

$$\text{prof}_Y(f^*F) \geq 3.$$

PROOF. One reduces, as in 1.14 and 1.15, to the case where X and S are strictly local schemes, f a local homomorphism and Y the closed point x of X . Let $s = f(x)$ be the closed point of S ; one has the commutative diagram:

$$\begin{array}{ccccc} X - X_s & \xrightarrow{i} & X - \{x\} & \xrightarrow{j} & X \\ \downarrow h & & \searrow g & & \swarrow f \\ S - \{s\} & \xrightarrow{k} & S & & \end{array}$$

1°) a) *Case $n = 1$.*

If one has $\text{prof}_s(F) \geq 1$, then the morphism $F \rightarrow k_*k^*F$ is injective, hence the morphism $f^*F \rightarrow f^*(k_*k^*F)$ is also injective. On the other hand it follows from the fact that f is locally (-1) -acyclic, that the morphism $f^*(k_*k^*F) \rightarrow (j.i) * (f^*F|_{X-X_s})$ is injective. Finally, the composite morphism $f^*F \rightarrow (j.i) * (f^*F|_{X-X_s})$ is injective, which shows that one has $\text{prof}_{X_s}(f^*F) \geq 1$, hence also $\text{prof}_x(f^*F) \geq 1$.

If one has $\text{prof}_x(X_s) \geq 1$, one considers the commutative diagram

$$(*) \quad \begin{array}{ccc} H^0(X, f^*F) & \xrightarrow{v} & H^0(X - \{x\}, f^*F) \\ \wr \downarrow & & \downarrow \\ H^0(X_s, f^*F) & \xrightarrow{v'} & H^0(X - \{x\}, f^*F); \end{array}$$

By hypothesis, v' is injective hence so is v .

b) *Case $n = 2$.* One considers the commutative diagram

$$(**) \quad \begin{array}{ccccc} H^0(S, F) & \xrightarrow{u} & H^0(S - \{s\}, F) & & \\ m \downarrow \wr & & n \downarrow \wr & & \\ H^0(X, f^*F) & \xrightarrow{v} & H^0(X - \{x\}, f^*F) & \xrightarrow{w} & H^0(X - X_s, f^*F); \end{array}$$

one must show that v is bijective. The morphism m is evidently bijective, and, as f is 0-acyclic, n is also bijective.

If one has $\text{prof}_s(F) \geq 2$, u is bijective. As one has seen in a), the sole hypothesis $\text{prof}_s(F) \geq 1$ implies the relation $\text{prof}_{X_s}(f^*F) \geq 1$; consequently v and w are injective; it then follows from (***) that v is bijective.

If one has $\text{prof}_x(X_s) \geq 2$, then g is 0-acyclic (for it is locally 0-acyclic and its fibres are 0-acyclic). It follows that $v \cdot m$ is bijective, hence v is bijective.

If one has $\text{prof}_s(F) \geq 1$ and $\text{prof}_x(X_s) \geq 1$, then one already knows that v and w are injective. Let z be a maximal point of $X_s - \{x\}$ (such a point exists by the hypothesis $\text{prof}_x(X_s) \geq 1$), $\bar{Z}$ the strict localization of X at a geometric point over z and $\overline{f^*F}$ the inverse image of f^*F on $\bar{Z}$. Consider the commutative diagram

$$\begin{array}{ccccc}
 H^0(S, F) & \xrightarrow{\quad} & H^0(S - \{s\}, F) & & \\
 \swarrow m \sim & & \searrow n \sim & & \\
 H^0(X, f^*F) & \xrightarrow{v} & H^0(X - \{x\}, f^*F) & \xrightarrow{w} & H^0(X - X_s, f^*F) \\
 \searrow m' \sim & & \swarrow r & & \searrow n' \sim \\
 & & H^0(\bar{Z}, f^*F) & \xrightarrow{\quad} & H^0(\bar{Z} - \{\bar{z}\}, \overline{f^*F})
 \end{array}$$

the morphism $m' \cdot m$ is obviously bijective and it follows from the fact that f is locally 0-acyclic that $n' \cdot n$ is bijective; consequently m' and n' are also bijective. As w is injective, r is also injective and consequently v is bijective.

2°) Taking b) into account, one already knows that $\text{prof}_x(f^*F) \geq 2$.

If one has $\text{prof}_s(F) \geq 3$, then $R^1 k_*(k^*F) = 1^6$. As f is locally 1-aspherical, one has $R^1(j \cdot i)_*(f^*F|_{X-X_s}) = f^*(R^1 k_*(k^*F)) = 1$. One therefore has $\text{prof}_{X_s}(f^*F) \geq 3$ and consequently one has $\text{prof}_x(f^*F) \geq 3$.

If one has $\text{prof}_x(X_s) \geq 3$, then g is 1-aspherical (for g is locally 1-aspherical and its fibres are 1-aspherical). One therefore has $H^1(X - \{x\}, f^*F) = H^1(S, F) = 1$ and consequently $\text{prof}_x(f^*F) \geq 3$.

If one has $\text{prof}_s(F) \geq 2$ and $\text{prof}_x(X_s) \geq 1$, one uses the exact sequence (SGA 4 XII 3.2):

$$1 \longrightarrow R^1 j_*(i_*(f^*F|_{X-X_s})) \longrightarrow R^1(j \cdot i)_*(f^*F|_{X-X_s}) \longrightarrow j_*(R^1 i_*(f^*F|_{X-X_s})).$$

As f and g are locally 1-aspherical, one has

$$\begin{aligned}
 R^1(j \cdot i)_*(f^*F|_{X-X_s}) &\simeq f^*(R^1 k_*(k^*F)) \\
 R^1 i_*(f^*F|_{X-X_s}) &\simeq g^*(R^1 k_*(k^*F));
 \end{aligned}$$

the preceding exact sequence is then written in the form

$$(***) \quad 1 \longrightarrow R^1 j_*(i_*(f^*F|_{X-X_s})) \longrightarrow f^*(R^1 k_*(k^*F)) \xrightarrow{a} j_*(j^*(f^*(R^1 k_*(k^*F)))).$$

The hypothesis $\text{prof}_s(F) \geq 2$ shows that the morphism $F \rightarrow k_*k^*F$ is bijective; applying g^* , one finds, taking into account the fact that g is locally 0-acyclic, $f^*F|_{X-\{x\}} = i_*(f^*F|_{X-X_s})$. The hypothesis $\text{prof}_x(X_s) \geq 1$ shows that the morphism a is injective (note that $f^*(R^1 k_*(k^*F))$ is a sheaf equal to 1 outside X_s and constant on X_s). It then follows from (***) that one has $R^1 j_*(f^*F|_{X-\{x\}}) = 1$, hence $\text{prof}_x(f^*F) \geq 3$.

If one has $\text{prof}_s(F) \geq 1$ and $\text{prof}_x(f^*F) \geq 2$, one considers the sheaf of homogeneous spaces G defined by the exact sequence

$$1 \longrightarrow F \longrightarrow k_*k^*F \longrightarrow G \longrightarrow 1.$$

⁶N.D.E. the trivial torsor is successively denoted 0 or 1 in what follows; one has left this double notation, which, on reflection, introduces no ambiguity.

Applying to this exact sequence the exact functor g^* and using (SGA 4 XII 3.1), one obtains the following commutative diagram whose rows are exact:

$$\begin{array}{ccccccc} f^*(k_*k^*F) & \longrightarrow & f^*G & \longrightarrow & 1 \\ \downarrow & & \downarrow b & & \\ j_*(g^*(k_*k^*F)) & \xrightarrow{u} & j_*(g^*G) & \longrightarrow & R^1 j_*(g^*F) & \longrightarrow & R^1 j_*(g^*(k_*k^*F)). \end{array}$$

As $\text{prof}_x(X_s) \geq 2$, the morphism b is bijective hence u is surjective and one thus has a map with kernel reduced to the identity element:

$$1 \longrightarrow R^1 j_*(g^*F) \longrightarrow R^1 j_*(g^*(k_*k^*F)) = R.$$

As $g^*(k_*k^*F) \simeq i_*(f^*F|_{X-X_s})$ (for g is locally 0-acyclic), R is identified with the first term of the exact sequence (**); now one has seen in the preceding case that $R = 1$ as soon as one has $\text{prof}_x(X_s) \geq 1$, which proves that $\text{prof}_x(f^*F) \geq 3$ and completes the proof of 1.18. $\square$

The corollaries which follow are generalizations of (SGA 4 XVI 3.2 and 3.3).

COROLLARY 1.19. *Let $f : X \rightarrow S$ be a flat morphism, with separable fibres, of locally noetherian schemes and Y a closed subset of X . Suppose that for every point $s \in f(Y)$, the fibre Y_s is rare⁷ in X_s and that one of the following two conditions is satisfied:*

- a) *the closure of $f(Y)$ is rare in S .*
- b) *X_s is geometrically unibranch at the points of Y_s .*

Then one has

$$\text{prof}_Y(X) \geq 2.$$

Indeed it follows from the hypothesis made on f that f is locally 0-acyclic (SGA 4 XV 4.1). One then applies 1.13. The hypothesis that Y_s is rare in X_s (resp. $\overline{f(Y)}$ rare in S) is equivalent by 1.6 b) to the relation $\text{prof}_{Y_s}(X_s) \geq 1$ (resp. $\text{prof}_{\overline{f(Y)}}(S) \geq 1$). The hypothesis that X_s is geometrically unibranch at each point of Y_s is equivalent to saying that the strict localization of X_s at a geometric point of Y_s is irreducible; knowing that Y_s is rare in X_s , this obviously implies $\text{prof}_{Y_s}(X_s) \geq 2$, thanks to 1.8. In either case 1.13 indeed gives $\text{prof}_Y(X) \geq 2$.

COROLLARY 1.20. *Let $f : X \rightarrow S$ be a regular morphism (EGA IV 6.8.1) of locally noetherian schemes, Y a closed subset of X . Suppose that, for every point $s \in f(Y)$, one of the following conditions is realized:*

- a) *One has $\text{codim}(Y_s, X_s) \geq 2$.*
- b) *One has $\text{codim}(Y_s, X_s) \geq 1$ and $\text{prof}_s(S) \geq 1$.*
- c) *One has $\text{prof hop}_s(S) \geq 3$.*

Then one has

$$\text{prof hop}_Y(X) \geq 3.$$

Indeed this follows from 1.18, given that hypothesis a) implies $\text{prof hop}_{Y_s}(X_s) \geq 3$ (cf. 1.11), and that the condition $\text{codim}(Y_s, X_s) \geq 1$ obviously implies $\text{prof}_Y(X) \geq 2$.

2. Technical lemmas

2.1. Let S be a locally noetherian scheme, $f : X \rightarrow S$ a morphism locally of finite type, t a point of S . If $x \in X$ is such that $s = f(x) \in \text{Spec } \mathcal{O}_{S,t}$, one sets

$$\delta_t(x) = \deg \text{tr } k(x)/k(s) + \dim(\overline{\{s\}}),$$

where $\overline{\{s\}}$ denotes the closure of s in $\text{Spec } \mathcal{O}_{S,t}$, $k(x)$ and $k(s)$ the residue fields of x and s respectively. If S is a local ring with closed point t , one also writes $\delta(x)$ instead of $\delta_t(x)$ (cf. SGA 4 XIV 2.2).

⁷**N.D.E.** “rare” = “of empty interior”, cf. Bourbaki TG IX.52.

LEMMA 2.1.1. *Let there be given a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

where S and S' are noetherian local rings, with closed points t and t' respectively, g a faithfully flat morphism such that $g^{-1}(t) = t'$, f a morphism locally of finite type. Let $x' \in X'$, $x = h(x')$, $s = f(x)$, $s' = f'(x')$; then one has

$$\delta(x') \leq \delta(x).$$

Moreover the preceding inequality is an equality if and only if one has:

$$\deg \operatorname{tr} k(x)/k(s) = \deg \operatorname{tr} k(x')/k(s') \text{ and } \dim(\overline{\{s\}}) = \dim(\overline{\{s'\}}).$$

In particular, given $x \in X$, one can find x' such that one has $\delta(x) = \delta(x')$.

Indeed, one has (EGA IV 6.11)

$$\dim(\overline{\{s\}}) = \dim g^{-1}(\overline{\{s\}}).$$

It follows that, for every point s' of $g^{-1}(s)$, one has the relation $\dim(\overline{\{s'\}}) \leq \dim(\overline{\{s\}})$, and that, given s , one can find $s' \in g^{-1}(s)$ such that one has equality. Let then Z denote the schematic closure of x in the fibre X_s of X at s , and let $Z' = Z \times_{\operatorname{Spec} k(s)} \operatorname{Spec} k(s')$. Then Z' is equidimensional of dimension $\deg \operatorname{tr} k(x)/k(s)$; one thus has, for every point $x' \in Z'_x$,

$$\deg \operatorname{tr} k(x')/k(s') \leq \deg \operatorname{tr} k(x)/k(s), \text{ and one has equality}$$

when x' is a maximal point of Z'_x . Whence immediately the announced conclusion.

2.2. Let $f : X \rightarrow S$ be a morphism locally of finite type and T a closed subset of S . Let $x \in X$, $s = f(x)$; we shall set

$$\delta_T(x) = \deg \operatorname{tr} k(x)/k(s) + \operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) = \inf_{t \in T \cap \overline{\{s\}}} \delta_t(x).$$

LEMMA 2.2.1. *Let there be given a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

where the schemes S and S' are locally noetherian, catenary, the morphism f locally of finite type and g faithfully flat. Let T be a closed subset of S , T' a closed subset of S' , such that $g(T') \subset T$, x' an element of X' , $x = h(x')$ and

$$h_{x'} : \operatorname{Spec} \mathcal{O}_{X',x'} \longrightarrow \operatorname{Spec} \mathcal{O}_{X,x}$$

the morphism induced by h . Then one has:

$$\delta_T(x) - \delta_{T'}(x') \leq \dim h_{x'}^{-1}(x).$$

Let $s' = f'(x')$, $s = f(x)$. One has, by definition:

$$\begin{aligned} \delta_T(x) - \delta_{T'}(x') &= \deg \operatorname{tr} k(x)/k(s) - \deg \operatorname{tr} k(x')/k(s') \\ &\quad + \operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) - \operatorname{codim}(\overline{\{s'\}} \cap T', \overline{\{s'\}}). \end{aligned}$$

Since g is faithfully flat, it follows from (EGA IV 6.1.4) that one has

$$\begin{aligned} (*) \quad \operatorname{codim}(\overline{\{s\}} \cap T, \overline{\{s\}}) &= \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap g^{-1}(T), g^{-1}(\overline{\{s\}})) \\ &\leq \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap T', g^{-1}(\overline{\{s\}})); \end{aligned}$$

since S' is catenary, one has, by (EGA 0_{IV} 14.3.2 b)):

$$\begin{aligned} \operatorname{codim}(\overline{\{s'\}} \cap T', g^{-1}(\overline{\{s\}})) &= \operatorname{codim}(\overline{\{s'\}} \cap T', \overline{\{s'\}}) + \operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}})) \\ &= \operatorname{codim}(\overline{\{s'\}} \cap T', g^{-1}(\overline{\{s\}}) \cap T') + \operatorname{codim}(g^{-1}(\overline{\{s\}}) \cap T', g^{-1}(\overline{\{s\}})). \end{aligned}$$

One deduces from this relation and from (*):

$$\delta_T(x) - \delta_{T'}(x') \leq \deg \operatorname{tr} k(x)/k(s) - \deg \operatorname{tr} k(x')/k(s') + \operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}})).$$

Let us compute $\operatorname{codim}(\overline{\{s'\}}, g^{-1}(\overline{\{s\}})) = \dim \mathcal{O}_{S'_s, s'}$ (where S'_s is the fibre of S' at s). Let Z be the closed image of x in X_s and $Z' \subset X'_s$ the scheme defined by the cartesian square

$$\begin{array}{ccc} Z' & \longrightarrow & Z \\ \downarrow & & \downarrow \\ S'_s & \longrightarrow & \operatorname{Spec} k(s) \end{array}.$$

The morphism $Z \rightarrow \operatorname{Spec} k(s)$ is flat, locally of finite type and one has $\dim Z = \deg \operatorname{tr} k(x)/k(s)$. It then follows from (EGA IV 6.1.2) that

$$\dim(\mathcal{O}_{Z', x'}) = \dim(\mathcal{O}_{S'_s, s'}) + \deg \operatorname{tr} k(x)/k(s) - \deg \operatorname{tr} k(x')/k(s');$$

taking into account the fact that $Z'_s \simeq Z \otimes_{k(s)} k(s')$, one then obtains:

$$\delta_T(x) - \delta_{T'}(x') \leq \dim(\mathcal{O}_{Z', x'}).$$

Now $\operatorname{Spec}(\mathcal{O}_{Z', x'})$ identifies with the fibre at x of the morphism

$$(\operatorname{Spec}(\mathcal{O}_{X', x'}))_s \longrightarrow (\operatorname{Spec}(\mathcal{O}_{X, x}))_s,$$

hence also with the fibre at x of $h_{x'}$, which proves the theorem.

2.3. The proofs of the theorems of no. 4 are based on duality theory; they use the lemmas which follow. Let m be an integer which is a power of a prime number ℓ ; if X is a scheme, all sheaves considered on X are sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules; one then has the notion of dualizing complex on X (SGA 5 I 1.7). Suppose that such a complex K exists on X ; then, for each geometric point $\bar{x}$ above a point x of X , one deduces from K (cf. SGA 5 I 4.5) a dualizing complex $K_{\bar{x}}$ on $\operatorname{Spec} k(\bar{x})$, so that one has $K_{\bar{x}} \simeq \mathbf{Z}/m\mathbf{Z}[n]$ (the brackets denoting the translation functor) for a certain integer n depending only on x . We shall set

$$\delta_K(x) = n.$$

If K is normalized at the point x (SGA 5 I 4.5), one thus has $n = 0$.

LEMMA 2.3.1. *Let X be a locally noetherian scheme, equipped with a dualizing complex K . If x and x' are two points of X , such that x is a specialization of x' and that one has $\operatorname{codim}(\overline{\{x\}}, \overline{\{x'\}}) = 1$, then one has*

$$\delta^K(x) = \delta^K(x') - 2.$$

One can first reduce to the case where X is a strictly local scheme. Indeed, let $\bar{X}$ be the strict localization of X at a geometric point $\bar{x}$ above x , $i : \bar{X} \rightarrow X$ the canonical morphism, $\bar{x}'$ a geometric point of $\bar{X}$ above x' . Then i^*K is a dualizing complex on $\bar{X}$ and one has (SGA 5 I 4.5)

$$(i^*K)_{\bar{x}} \simeq K_{\bar{x}} \text{ and } (i^*K)_{\bar{x}'} \simeq K_{\bar{x}'},$$

which completes the reduction to the strictly local case.

If $j : \overline{\{x'\}} \rightarrow X$ denotes the immersion of the reduced closed subscheme of X , with underlying space $\overline{\{x'\}}$, then $R^!j(K)$ is a dualizing complex on $\overline{\{x'\}}$ and one sees at once, using (SGA 5 I 4.5), that it suffices to prove the lemma for $\overline{\{x'\}}$. One is thus reduced to the case where X is a strictly local integral scheme of dimension 1.

Let then X' be the normalization of X and $f : X' \rightarrow X$ the canonical morphism; f is an integral, surjective, radicial morphism, and it follows that f^*K is a dualizing complex on

X' and that it suffices to prove the lemma for X' and for the points above x and x' . One is thus reduced to the case where X is a local, integral, regular scheme of dimension 1, but one knows (cf. SGA 5 I 4.6.2 and 5.1) that then $\mu_m[2]$ and $\mathbf{Z}/m\mathbf{Z}$ are dualizing complexes, normalized respectively at the points x and x' ; the lemma follows at once.

LEMMA 2.3.2. *Let S be a noetherian local scheme, $f : X \rightarrow S$ a morphism of finite type. If K is a dualizing complex on S , normalized at the closed point t of S and if $R^!f(K) = K'$ is a dualizing complex on X (cf. SGA 5 I 3.4.3), one has, for every point x of X :*

$$\delta^{K'}(x) = 2\delta(x).$$

Indeed, let $s = f(x)$ and x' a closed point of the fibre X_s ; then one has $\delta^{K'}(x') = \delta^K(s)$ and by 2.3.1

$$\delta^K(s) = 2 \operatorname{codim}(\overline{\{t\}}, \overline{\{s\}}) = 2 \dim(\overline{\{s\}}).$$

As one can choose for x' a specialization of x , one has by 2.3.1

$$\delta^{K'}(x) = \delta^{K'}(x') + 2 \operatorname{codim}(\overline{\{x\}}, \overline{\{x'\}}) = \delta^{K'}(x') + 2 \deg \operatorname{tr} k(x)/k(s);$$

the lemma follows at once.

The following lemma will serve only for the converse of the Lefschetz theorem, in no. 4:

LEMMA 2.3.3. *Let there be given a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array},$$

where S is a strictly local excellent scheme of characteristic zero, S' the completion of S and f a morphism of finite type. Let ℓ be a prime number, $x \in X$, Z the schematic closure of X'_x in X' , and $i : X'_x \rightarrow Z$, $j : Z \rightarrow X$ the canonical morphisms. Then, if $k : X' \rightarrow R$ is a closed immersion of X' into a regular excellent scheme R , of characteristic zero, the complex

$$K' = i^*(R^!(k.j))(\mathbf{Z}/\ell\mathbf{Z})$$

is a constant dualizing complex on X'_x (i.e. having a unique non-zero cohomology sheaf, isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$).

Taking (SGA 5 I 3.4.3) into account, the only thing to prove is that K' is constant. Now, since Z is excellent, the set of points of Z whose local rings are regular is an open set U (EGA IV 7.8.3 (iv)), and U obviously contains X'_x which is regular. Let then

$$u : U \longrightarrow R$$

be the canonical immersion of U into R ; it follows from the purity theorem (SGA 4 XIX 3.2 and 3.4) and from the isomorphism

$$(\mu_l)_S \simeq (\mathbf{Z}/\ell\mathbf{Z})_S$$

(S strictly local) that one has

$$R^!u(\mathbf{Z}/\ell\mathbf{Z}) \simeq \mathbf{Z}/\ell\mathbf{Z}[2c],$$

where c is a locally constant function on U , necessarily constant in a neighbourhood of X'_x , since the fibres of g are geometrically integral by (EGA IV 18.9.1), hence X'_x is integral. The lemma follows at once.

3. Converse of the affine Lefschetz theorem

The present no. will be used in no. 4 to prove a converse to the ‘‘Lefschetz theorem’’; a reader who is interested only in the direct part of that theorem may therefore omit the reading of the present no.

3.1. Let us recall the statement of the affine Lefschetz theorem⁸ (SGA 4 XIX 6.1 bis):

Let S be a strictly local excellent scheme of characteristic zero, $f : X \rightarrow S$ an affine morphism of finite type and F a torsion sheaf on X . Then, if one sets

$$\delta(F) = \sup\{\delta(x) | x \in X \text{ and } F_{\bar{x}} \neq 0\},$$

one has

$$H^q(X, F) = 0 \text{ for } q > \delta(F).$$

Before stating the converse, let us prove some lemmas.

LEMMA 3.2. *Let K be a field, ℓ a prime number distinct from the characteristic of K and F a sheaf of ℓ -torsion on K , constructible, non-zero. Suppose that the ℓ -cohomological dimension of K (SGA 4 X 1) is equal to n (this is realized for example if K is the fraction field of a strictly local excellent integral ring, of characteristic zero, of dimension n (SGA 4 XIX 6.3), or if K is a finitely generated extension of transcendence degree n of a separably closed field (SGA 4 X 2.1)). Then one can find a finite separable extension L of K such that one has:*

$$H^n(L, F|_L) \neq 0.$$

One can find a finite extension K' of K such that the restrictions of F and of μ_ℓ to $\text{Spec } K'$ are constant sheaves. One then has $\text{cd}_\ell(K') = \text{cd}_\ell(K) = n$ (SGA 4 X 2.1), and it follows from ([2] II §3 Prop. 4 (iii)) that one can find a finite extension L of K' such that one has

$$H^n(L, \mu_\ell) \neq 0, \quad \text{i.e. } H^n(L, \mathbf{Z}/\ell\mathbf{Z}) \neq 0.$$

Now the functor $H^n(L, \cdot)$ is right exact on the category of sheaves of ℓ -torsion, since $\text{cd}_\ell(L) = n$; as F admits a quotient isomorphic to $\mathbf{Z}/\ell\mathbf{Z}$, one also has $H^n(L, F|_L) \neq 0$.

COROLLARY 3.3. *Let k be a field, K a finitely generated extension of transcendence degree n of k , F a sheaf of ℓ -torsion on K , constructible, non-zero, with ℓ prime to the characteristic of k . Then one can find a finite separable extension L of K such that, if $u : \text{Spec } L \rightarrow \text{Spec } k$ denotes the canonical morphism, one has*

$$R^n u_* (F|_{\text{Spec } L}) \neq 0.$$

When the field k is separably closed, the corollary is a special case of 3.2. In the general case, one can find a finite separable extension k_1 of k such that the irreducible components of $K \otimes_k k_1$ are geometrically irreducible (EGA IV 4.5.11); let K_1 be one of them. If k' is a separable closure of k_1 , then $K' = K_1 \otimes_{k_1} k'$ is a field, and one has by (EGA IV 4.2)

$$\deg \text{tr } K'/k' = \deg \text{tr } K/k = n.$$

It then follows from 3.2 that one can find a finite separable extension L' of K' such that one has $H^n(L', F|_{L'}) \neq 0$. But one has $k' = \varinjlim_i k_i$, where k_i runs through the finite extensions of k_1 contained in k' , and consequently $K' = \varinjlim_i (k_i \otimes_{k_1} K_1)$. It follows that one can find an index i and a finite separable extension L of $k_i \otimes_{k_1} K_1 = K_i$, such that one has $L' \simeq L \otimes_{K_i} K'$. The extension L of K answers the question; indeed it follows from the commutative diagram

$$\begin{array}{ccc} & \text{Spec } L & \\ v \swarrow & & \searrow u \\ \text{Spec } k_i & \xrightarrow{w} & \text{Spec } k, \end{array}$$

⁸**N.D.E.** Gabber has proved the following generalization. Let Y be a strictly local scheme of arithmetic type over a regular noetherian scheme S of dimension ≤ 1 . Let $f : X \rightarrow Y$ be an affine morphism of finite type, $\Lambda = \mathbf{Z}/n\mathbf{Z}$ with n invertible on X and F a Λ -sheaf. Then $H^q(X, F) = 0$ if $q > \delta(F)$. One deduces from this the following local Lefschetz theorem. Let $\mathcal{O}$ be strictly local of arithmetic type over S . For every $f \in \mathcal{O}$ not a zero-divisor and every Λ -sheaf F on $\text{Spec } (\mathcal{O}[f^{-1}])$, one has $H^q(\text{Spec } (\mathcal{O}[f^{-1}]), F) = 0$ for $q > \dim(\mathcal{O})$. Cf. (Fujiwara K., “A Proof of the Absolute Purity Conjecture (after Gabber)”, in *Algebraic geometry 2000, Azumino (Hotaka)*, Adv. Stud. in Pure Math., vol. 36, 2002, p. 153-183, §5) and especially Illusie’s article (Illusie L., “Perversité et variation”, *Manuscripta Math.* **112** (2003), p. 271-295). This result is one of the crucial points used by Gabber in his proof of Grothendieck’s purity theorem (cf. the note (4), page 8).

with w finite hence $R^q w_* = 0$ if $q > 0$, that one has

$$R^n u_*(F|_{\text{Spec } L}) \simeq w_*(R^n v_*(F|_{\text{Spec } L})).$$

Now $R^n v_*(F|_{\text{Spec } L}) \neq 0$, since $H^n(L', F|_{L'}) \neq 0$; one therefore also has $R^n u_*(F|_{\text{Spec } L}) \neq 0$.

Let us recall the following well-known lemma (cf. EGA 0_{III} 10.3.1.2 and EGA IV 18.2.3):

LEMMA 3.4. *Let X be a scheme, x a point of X , K a finite separable extension of $k(x)$. Then there exists a scheme X_1 étale over X , affine, and a point $x_1 \in X_1$ above x , such that $k(x_1)$ is $k(x)$ -isomorphic to K .*

We shall use in no. 4 the following technical form of the converse of 3.1.

PROPOSITION 3.5. *Let a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S, \end{array}$$

where the schemes S and S' are strictly local excellent of characteristic zero, the morphism f locally of finite type, g regular (EGA IV 6.8.1) surjective, the closed fibre of g reduced to the closed point of S' . Given an S -scheme X_1 (resp. an S -morphism f_1 , etc.), we shall denote by X'_1 (resp. f'_1 , etc.) the scheme $X_1 \times_S S'$ (resp. the morphism $(f_1)_{(S')}$, etc.). Let F be a sheaf of $\mathbf{Z}/m\mathbf{Z}$ -modules on X' (m a power of a prime number ℓ), constructible, satisfying the following conditions:

- (i) For every point $x \in X$, one can find a finite separable extension K of $k(x)$ such that the restriction of F to the fibre $(X')_{(\text{Spec } K)}$ comes by inverse image from a constructible sheaf on $\text{Spec } K$.
- (ii) For every morphism $f_1 : X_1 \rightarrow S$, with X_1 étale over X , affine, for every point $s \in S$ and for every integer $q > 0$, one can find a finite separable extension K of $k(s)$ such that the restriction of $R^q f_{1*}(F|_{X'_1})$ to the fibre $S'_{(\text{Spec } K)}$ comes by inverse image from a constructible sheaf on $\text{Spec } K$.

Let n be an integer, and suppose that for every scheme X_1 étale over X , affine, one has

$$H^i(X'_1, F) = 0 \text{ for } i > n.$$

Then, if $\bar{x}'$ is a geometric point above the point $x' \in X'$ such that $F_{\bar{x}'} \neq 0$, one has

$$\delta(x') \leq n.$$

Let Z' be the set of points x' of X' such that one has $F_{\bar{x}'} = 0$. Then, if $Z = h(Z')$, one has by (i) $Z' = h^{-1}(Z)$; let $x' \in X'$, $x = h(x')$, $s' = f'(x')$, $s = f(x)$. It follows from 2.1.1 and from the fact that the function δ decreases under specialization, that it suffices to prove the inequality $\delta(x') \leq n$ when x is a maximal point of Z and x' is such that one has

$$r = \deg \text{tr } k(x)/k(s) = \deg \text{tr } k(x')/k(s') \text{ and } d = \dim \overline{\{s\}} = \dim \overline{\{s'\}}$$

Let x' be such a point; it suffices to show that one can find a scheme X_1 étale over X , affine, such that one has

$$H^{d+r}(X'_1, F) \neq 0.$$

The set Z' is constructible (SGA 4 IX 2.4), hence the same is true of Z (EGA IV 1.9.12); one may then suppose, after possibly restricting X to a neighbourhood of x , that Z is an irreducible closed subset with generic point x . Let $T = f(Z)$; T is a constructible set contained in $\overline{\{s\}}$; one can therefore find an affine open U of S such that $s \in U$ and that $T \cap U = T_U$ is an irreducible closed subset of U with generic point s .

Then let V be a scheme étale over X , affine, whose image in X contains x and whose image in S is contained in U ; let Z_V be the inverse image of Z in V and $u : Z_V \rightarrow T_U$ the canonical morphism. Let W be a scheme étale over U , affine, one then denotes by T_W the

inverse image of T_U in W and let $X_1 = W \times_U V$. As F is zero outside Z' , one has the spectral sequence

$$E_2^{pq} = H^p((T_W)', R^q u'_*(F|_{(Z_V)'})) \implies H^*(X'_1, F).$$

We shall show that one can choose V and W in such a way that one has

- a) $E_2^{pq} = 0$ for $p > d$ and for $q > r$.
- b) $E_2^{dr} \neq 0$.

The relation $H^{d+r}(X'_1, F) \neq 0$ will then follow from the spectral sequence.

1°) Set $G_q = R^q u'_*(F|_{(Z_V)'});$ then one has:

$$(G_q)_{\bar{s}'} = H^q((Z_V)'_{\bar{s}'}, F|_{(Z_V)'_{\bar{s}'}},$$

since s' is a maximal point of $(T_U)'$. As the fibre $(Z_V)'_{\bar{s}'}$ is an affine scheme of finite type of dimension r over a separably closed field, it follows from 3.1 that one has

$$(G_q)_{\bar{s}'} = 0 \text{ for } q > r.$$

For $q > r$, let Y'_q be the set of points of $(T_U)'$ where the geometric fibre of G_q is $\neq 0$ and $Y_q = g(Y'_q)$; then one has $Y'_q = g^{-1}(Y_q)$ by (ii), hence Y_q is a constructible subset of T_U (SGA 4 XIX 5.1 and EGA 1.9.12) which does not contain s ; after possibly restricting U to an open neighbourhood of s , one may suppose that one has $G_q = 0$ for $q > r$, hence $E_2^{pq} = 0$ for $q > r$.

Moreover, as $(T_W)'$ is an affine scheme of finite type over $g^{-1}(\overline{\{s\}})$, one has for any q (cf. 3.1):

$$H^p((T_W)', G_q) = 0 \text{ for } p > \dim g^{-1}(\overline{\{s\}}) = d,$$

whence condition a).

2°) Let us show that one can choose V so that one has $(G_r)_{\bar{s}'} \neq 0$. By (i), there exists a constructible sheaf I , defined over a finite separable extension K of $k(x)$, whose inverse image on $(X')_{(\text{Spec } K)}$ is isomorphic to $F|_{(X')_{(\text{Spec } K)}}$. By 3.3, one finds a finite separable extension L of K such that, if $v : \text{Spec } L \rightarrow \text{Spec } k(s)$ denotes the canonical morphism, one has $R^r v_*(I) \neq 0$. As the morphism $S'_s \rightarrow \text{Spec } k(s)$ is regular, one has by (SGA 4 XIX 4.2):

$$R^r v'_*(F|_{(\text{Spec } L)'}) \simeq (R^r v_*(I))' \neq 0.$$

By lemma 3.4, one can find a scheme X_2 étale over X , affine, and a point x_2 of X_2 above x , such that L is $k(x)$ -isomorphic to $k(x_2)$ and one may suppose X_2 over U . As x is a maximal point of Z , one has

$$\text{Spec } L \simeq \varprojlim_V Z_V,$$

where V runs through the affine open neighbourhoods of x_2 . One deduces from this by passage to the limit (SGA 4 VII 5.8), after restriction to the geometric fibre at $\bar{s}'$:

$$(R^r v'_*(F|_{\text{Spec } L})')_{\bar{s}'} = \varinjlim_V (R^r u'_*(F|_{(Z_V)'})_{\bar{s}'}),$$

which indeed shows that one can find V such that one has $(G_r)_{\bar{s}'} \neq 0$.

3°) The scheme V having been chosen in 2°), let us show that one can choose the scheme W so that one has

$$E_2^{dr} = H^d((T_W)', G_r) \neq 0.$$

By (ii), there exists a constructible sheaf J , defined over a finite separable extension K of $k(s)$, whose inverse image on $(S')_{(\text{Spec } K)}$ is isomorphic to $G_r|_{(S')_{(\text{Spec } K)}}$. By lemma 3.2, one can find a finite separable extension L of K such that one has

$H^d(\mathrm{Spec} L, J) \neq 0$. As the morphism $(S')_{(\mathrm{Spec} L)} \rightarrow \mathrm{Spec} L$ is acyclic (SGA 4 XIX 4.1 and XV 1.10 and 1.16), one has

$$H^d((\mathrm{Spec} L)', G_{r|(S')_{(\mathrm{Spec} L)'}}) = H^d(\mathrm{Spec} L, J) \neq 0.$$

By 3.4, one can find a scheme U_1 étale over U , affine, and a point s_1 above s , such that $k(s_1)$ is $k(s)$ -isomorphic to L . Now, s being a maximal point of T_U , one has

$$\mathrm{Spec} L \simeq \varprojlim_W T_W,$$

where W runs through the affine open neighbourhoods of s_1 . One deduces from this that $(\mathrm{Spec} L)' \simeq \varprojlim_W (T_W)'$, and by passage to the limit (SGA 4 VII 5.8):

$$H^d((\mathrm{Spec} L)', G_{r|(\mathrm{Spec} L)'}) \simeq \varinjlim_W H^d((T_W)', G_{r|(T_W)'});$$

hence one can find W such that one has

$$H^d((T_W)', G_{r|(T_W)'}) \neq 0,$$

which completes the proof of the theorem.

COROLLARY 3.6. *The hypotheses concerning S, S', f, f', m are those of 3.5. Let us now denote by F a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules on X' , with degrees bounded below and with constructible cohomology, and whose cohomology sheaves satisfy conditions (i) and (ii) of 3.5. Let n be an integer, and suppose that, for every scheme X_1 étale over X , affine, one has*

$$H^i(X'_1, F) = 0 \text{ for } i > n.$$

Then, if $\bar{x}'$ is a geometric point above a point x' of X' such that one has, for an integer j , $(\underline{H}^j(F))_{\bar{x}'} \neq 0$, one has

$$\delta(x') \leq n - j.$$

Let T' be the set of points of X' where the conclusion of 3.7 fails and suppose $T' \neq \emptyset$; let $T = f(T')$, x a maximal point of T and x' a point of X' above x . Let j be the largest integer such that one has $(\underline{H}^j(F))_{\bar{x}'} \neq 0$; one thus has $r = \delta(x) > n - j$. Let Z'_q be the set of points where the geometric fibre of $\underline{H}^q(F)$ is $= 0$ and $Z_q = h(Z'_q)$; one sees as in the proof of 3.5 that Z_q is constructible. One has obviously $Z'_q = \emptyset$ for $q > n$ and for q sufficiently small. The other values of q fall into three subsets. Let

$$Q_1 = \{q \mid x \in Z_q \text{ and a generalization of } x, \text{ distinct from } x, \notin Z_q\}.$$

One has $j \in Q_1$ and one can find an affine open neighbourhood U_1 of x such that, for every $q \in Q_1$, $U_1 \cap Z_q$ is an irreducible closed subset with generic point x . If $q \in Q_1$, one has

$$(*) \quad \delta(\underline{H}^q(F)|_{U'_1}) = \delta(x) \quad (\text{for the definition of } \delta(\underline{H}^q(F)) \text{ cf. 3.1}).$$

Let

$$Q_2 = \{q \mid \text{no generalization of } x \text{ belongs to } Z_q\}.$$

Then, if $j < q \leq n$, one has $q \in Q_2$, and one can find an affine open neighbourhood U_2 of x such that, for every $q \in Q_2$, one has $Z_q \cap U_2 = \emptyset$; one thus has

$$(**) \quad \underline{H}^q(F)|_{U'_2} = 0 \text{ for } q \in Q_2.$$

Finally, let

$$Q_3 = \{q \mid Z_q \text{ contains strict generalizations of } x\}.$$

Then one can find an affine open neighbourhood of x , U_3 , such that, for every $q \in Q_3$, all the maximal points of $Z_q \cap U_3$ are generalizations of x . If $q \in Q_3$, one has

$$(***) \quad \delta(\underline{H}^q(F)|_{U'_3}) \leq n - q.$$

For every scheme X_1 étale over $U_1 \cap U_2 \cap U_3$, affine, consider the hypercohomology spectral sequence

$$E_2^{pq} = H^p(X'_1, \underline{H}^q(F)) \implies H^*(X'_1, F).$$

One has $E_2^{pq} = 0$ for $q \in Q_2$ by (**). One has $E_2^{pq} = 0$ for $p + q \geq r + j$ except perhaps for $p = r, q = j$. Indeed this is clear if $q \in Q_2$; if $q \in Q_1$, one then has $p > r$ except if $p = r, q = j$ and this follows from 3.1 taking into account (*); finally if $q \in Q_3$, as $r > n - j$, one has $p > n - q$ and the assertion follows from 3.1 taking into account (***). Since $H^{r+j}(X'_1, F) = 0$, it follows from the spectral sequence that one has

$$H^r(X'_1, \underline{H}^j(F)) = 0;$$

now this implies, by 3.5, $\delta(x) < r$, which is absurd.

COROLLARY 3.7. *Let S be a strictly local excellent scheme of characteristic zero, $f : X \rightarrow S$ a morphism locally of finite type, m a power of a prime number, F a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules on X , bounded below, with constructible cohomology, and n an integer. Then the following conditions are equivalent:*

(i) *For every scheme X_1 étale over X , affine, one has*

$$H^i(X_1, F) = 0 \text{ for } i > n.$$

(ii) *For every geometric point $\bar{x}$ above the point x of X , and for every integer j such that one has $(\underline{H}^j(F))_{\bar{x}} \neq 0$, one has*

$$\delta(x) \leq n - j.$$

(i) $\Rightarrow$ (ii) is the special case of 3.6 obtained by setting $S = S'$.

(ii) $\Rightarrow$ (i) follows immediately from 3.1, using the hypercohomology spectral sequence

$$H^p(X_1, \underline{H}^q(F)) \Longrightarrow H^*(X_1, F).$$

4. Main theorem and variants

4.0. Let $g : X \rightarrow S$ be a separated morphism of finite type, T a closed subset of S , $Z = g^{-1}(T)$ and F a complex of abelian sheaves on X , bounded below in degrees. We call the i -th cohomology group of F , with proper support, with support in Z the group

$$H_{Z!}^i(X/S, F) = H_T^i(S, R_! g(F)),$$

where $R_! g$ denotes “direct image with proper support” (SGA 4 XVII). In the particular case where g is proper, one simply has

$$H_{Z!}^i(X/S, F) = H_Z^i(X, F).$$

PROPOSITION 4.1. *Let $f : U \rightarrow S$ be a morphism of finite type, F a complex of abelian sheaves on U , bounded below in degrees. Suppose that one has a factorization of f :*

$$\begin{array}{ccc} U & \xrightarrow{i} & X \\ & \searrow f & \swarrow g \\ & S & \end{array}$$

where i is an open immersion and g a separated morphism of finite type, and let G denote a complex of abelian sheaves on X , bounded below in degrees, which extends F . Let Y be a closed subscheme of X with underlying space $X - U$, so that one has a commutative diagram:

$$\begin{array}{ccc} Y & \xrightarrow{j} & X \\ & \searrow h & \swarrow g \\ & S & \end{array}$$

Finally let n be an integer and T a closed subset of S . Then the following conditions are equivalent:

(i) *One has $\text{prof}_T(R_! f(F)) \geq n$.*

(ii) *The canonical morphism*

$$\underline{H}_T^i(R_! g(G)) \longrightarrow \underline{H}_T^i(R_! h(j^* G))$$

is bijective for $i < n - 1$, injective for $i = n - 1$.

(iii) *For every scheme S' étale over S , if one denotes by X' (resp. f' , resp. etc.) the scheme $X \times_S S'$ (resp. the morphism $f_{(S')}$, resp. etc.), the canonical morphism*

$$H_{g'^{-1}(T')!}^i(X'/S', G') \longrightarrow H_{h'^{-1}(T')!}^i(Y'/S', j'^* G')$$

is bijective for $i < n - 1$, injective for $i = n - 1$.

Consider in the derived category $D^+(X)$ (cf. [3]) the distinguished triangle

$$\begin{array}{ccc} & j_* j^* G & \\ \swarrow & & \searrow \\ i_! F & \xrightarrow{\quad} & G. \end{array}$$

Applying to this triangle the functor $R_! g$, one obtains the triangle

$$(*) \quad \begin{array}{ccc} & R_! h(j^* G) & \\ \swarrow & & \searrow \\ R_! f(F) & \xrightarrow{\quad} & R_! g(G). \end{array}$$

Let us show that (i) $\Leftrightarrow$ (ii). Indeed, by definition 1.2, (i) is equivalent to the relation

$$\underline{H}_T^i(R_! f(F)) = 0 \quad \text{for } i < n;$$

now from (*) one deduces the exact sequence of sheaves

$$\longrightarrow \underline{H}_T^i(R_! f(F)) \longrightarrow \underline{H}_T^i(R_! g(G)) \longrightarrow \underline{H}_T^i(R_! h(j^* G)) \longrightarrow,$$

whence the equivalence of (i) and (ii).

(i) $\Leftrightarrow$ (iii). Indeed (i) is equivalent to saying that, for every scheme S' étale over S , one has the relation

$$(**) \quad H_{T'}^i(S', R_! f'(F')) = 0 \quad \text{for } i < n.$$

Now from (*) one deduces the exact sequence of abelian groups

$$\longrightarrow H_{T'}^i(S', R_! f'(F')) \longrightarrow H_{T'}^i(S', R_! g'(G')) \longrightarrow H_{T'}^i(S', R_! h'(j'^* G')) \longrightarrow ;$$

taking 4.0 into account, this exact sequence may be written in the form

$$\longrightarrow H_{T'}^i(S', R_! f'(F')) \longrightarrow H_{g'^{-1}(T')!}^i(X'/S', G') \longrightarrow H_{h'^{-1}(T')!}^i(Y'/S', j'^* G') \longrightarrow .$$

The equivalence of (i) and (iii) follows from this, taking into account the form (**) of (i).

4.2.0. When $f: U \rightarrow S$ is affine, we shall give *local* conditions on F in order that conditions (i) to (iii) of 4.1 be satisfied. In the sequel, the schemes considered are excellent schemes of characteristic zero, the sheaves are sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules, where m is a power of a prime number. If one had resolution of singularities in the sense of (SGA 4 XIX), the results stated, as well as their proofs, would still be valid for excellent schemes of equal characteristic, with m prime to the characteristic.

THEOREM 4.2. *Let S be an excellent scheme of characteristic zero and $f: U \rightarrow S$ a separated morphism of finite type. Let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules on U , bounded below in degrees and with constructible cohomology, n an integer and T a closed subset of S . Then the following conditions are equivalent:*

- (i) For every scheme U_1 étale over U , affine over S , one has, denoting by f_1 the structural morphism of U_1 and by F_1 the restriction of F to U_1 :

$$\mathrm{prof}_T(\mathrm{R}_! f_1(F_1)) \geq n$$

(cf. prop. 4.1 on the meaning of this relation).

- (ii) For every point u of U , one has:

$$\mathrm{prof}_u(F) \geq n - \delta_T(u),$$

where one sets (cf. 2.2): $\delta_T(u) = \deg \mathrm{tr}(k(x)/k(s)) + \mathrm{codim}(\overline{\{s\}} \cap T, \overline{\{s\}})$.

PROOF. 1°) Let t be a point of T , $\overline{S}$ the strict localization of S at a geometric point over t and S' the completion of $\overline{S}$, with closed point t' ; then $\overline{S}$ is excellent by (EGA IV 7.9.5), hence S' is a complete strictly local excellent scheme. Given a scheme U over X (resp. an S -morphism f , resp. etc.), we shall denote by U' (resp. f' , resp. etc.) the scheme $U \times_S S'$ (resp. the morphism $f_{(S')}$, resp. etc.). One has the cartesian square

$$\begin{array}{ccc} U' & \xrightarrow{h} & U \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S, \end{array}$$

in which the morphism g is regular (EGA IV 7.8.2). Let us show that it suffices to prove that (for every point $t \in T$) the following two properties are equivalent:

- (i) _{t} For every scheme U_1 étale over U , affine over S , setting $f_1: U_1 \rightarrow S$, one has

$$\mathrm{prof}_{t'}(\mathrm{R}_! f_1'(F_1')) \geq n.$$

- (ii) _{t} For every point u' of U' , one has

$$\mathrm{prof}_{u'}(F') \geq n - \delta_{t'}(u').$$

It suffices to prove the following lemma:

LEMMA 4.2.1. *One has (i) $\Leftrightarrow$ (i) _{t} for every $t \in T$ and (ii) $\Leftrightarrow$ (ii) _{t} for every $t \in T$.*

(i) $\Leftrightarrow$ (i) _{t} for every $t \in T$. Indeed (i) is equivalent to saying that, for every scheme U_1 étale over U , affine over S , one has

$$\mathrm{prof}_T(\mathrm{R}_! f_1(F_1)) \geq n;$$

now by 1.8

$$\mathrm{prof}_T(\mathrm{R}_! f_1(F_1)) = \inf_{t \in T} \mathrm{prof}_t(\mathrm{R}_! f_1(F_1)).$$

Since $g^*(\mathrm{R}_! f_1(F_1)) \simeq \mathrm{R}_! f_1'(F_1')$ (SGA 4 XVII), one has by 1.16

$$\mathrm{prof}_t(\mathrm{R}_! f_1(F_1)) = \mathrm{prof}_{t'}(\mathrm{R}_! f_1'(F_1')),$$

hence (i) is equivalent to saying that one has, for every $t \in T$, $\mathrm{prof}_{t'}(\mathrm{R}_! f_1'(F_1')) \geq n$, which is nothing other than (i) _{t} .

- (ii) _{t} for every $t \in T \Rightarrow$ (ii). Indeed let $u \in U$; one must show the relation

$$\mathrm{prof}_u(F) \geq n - \delta_T(u),$$

where $\delta_T(u) = \inf_{t \in T \cap \overline{\{s\}}} \delta_t(u)$ (cf. 2.2); one is thus reduced to showing that one has, for every $t \in T \cap \overline{\{s\}}$

$$\mathrm{prof}_u(F) \geq n - \delta_t(u).$$

Let u' be a point of U' such that one has $h(u') = u$ and $\delta_{t'}(u') = \delta_t(u)$ (cf. 2.1.1). Since h is locally acyclic (SGA 4 XIX 4.1), it follows from 1.16 and from the fact that u' is a generic point of U'_u that one has

$$\mathrm{prof}_{u'}(F') = \mathrm{prof}_u(F).$$

But by (ii) _{t} one has $\mathrm{prof}_u(F) = \mathrm{prof}_{u'}(F') \geq n - \delta_t(u)$, which proves (ii).

(ii) $\Rightarrow$ (ii) $_t$ for every t . With the notation of 2.2.1, for every point u' of U' , one has thanks to 1.16

$$\text{prof}_{u'}(F') \geq \text{prof}_u(F) + 2 \dim h_{u'}^{-1}(u) \geq \text{prof}_u(F) + \dim h_u^{-1}(u).$$

Taking 2.2.1 and (ii) into account, one obtains

$$\text{prof}_{u'}(F') \geq n - \delta_T(u) + \dim h_{u'}^{-1}(u) \geq n - \delta_{t'}(u'),$$

which is nothing other than (ii) $_t$.

2°) (ii) $_t \Leftrightarrow$ (i) $_t$. One immediately reduces to the case where F has bounded degrees, by truncating F at a sufficiently high degree. One can realize S' as a closed subscheme of a complete regular local scheme, hence excellent; it then follows from (SGA 5 I 3.4.3) that there exists a dualizing complex K on S' and that $R^1 f'(K) = K'$ is a dualizing complex on U' . We shall choose K in such a way that one has $\delta^K(t') = 0$ (for the definition of $\delta^K(t')$, cf. 2.3), and we shall denote by $D F'$ the dual of F' with respect to K' . One can reformulate the hypothesis (ii) $_t$ as follows:

LEMMA 4.2.2. *Let u' be a point of U' ; then the following conditions are equivalent:*

- (i) *One has $\text{prof}_{u'}(F') \geq n - \delta_{t'}(u')$.*
- (ii) *One has $(\underline{H}^q(D F'))_{\bar{u}'} = 0$ for $q > -n - \delta_{t'}(u')$ ($\bar{u}'$ a geometric point over u').*

Let $\bar{U}'$ be the strict localization of U' at $\bar{u}'$ and $\bar{F}'$ the inverse image of F by the morphism $\bar{U}' \rightarrow U'$. The relation $\text{prof}_{u'}(F') \geq n - \delta_{t'}(u')$ is equivalent by definition to the following:

$$(*) \quad H_{\bar{u}'}^i(\bar{F}') = 0 \quad \text{for } i > n - \delta_{t'}(u').$$

Let $D(H_{\bar{u}'}^i(\bar{F}'))$ be the dual of the abelian group $H_{\bar{u}'}^i(\bar{F}')$ with respect to $\mathbf{Z}/m\mathbf{Z}$. By 2.3.2, $K'[-2\delta_{t'}(u')] = K''$ satisfies $\delta^{K''}(u') = 0$; since F' has constructible cohomology, one has $\bar{D} F' = D(\bar{F}')$ and the local duality theorem (SGA 5 I 4.5.3) then shows that one has

$$D(H_{\bar{u}'}^i(\bar{F}')) \simeq (H^{-i-2\delta_{t'}(u')}(D F'))_{\bar{u}'}.$$

Thus $(*)$ is equivalent to the relation

$$(**) \quad (\underline{H}^q(D F'))_{\bar{u}'} = 0 \quad \text{for } q > -n - \delta_{t'}(u').$$

We are now in a position to prove the theorem. The relation (ii) $_t$ is equivalent to the relation $(**)$. Let $G^q = \underline{H}^q(D F')$; the affine Lefschetz theorem (3.1) implies in particular that, for every scheme U_1 étale over U , affine over S , one has

$$H^p(U_1', G^q) = 0 \quad \text{for } p > \delta(G^q),$$

where $\delta(G^q)$ is the upper bound of the $\delta_{t'}(u')$ for those u' such that one has $G_{\bar{u}'}^q \neq 0$; by $(**)$ one has $\delta(G^q) \leq -n - q$, hence (ii) $_t$ implies the relation

$$H^p(U_1', \underline{H}^q(D F')) = 0 \quad \text{for } p > -q - n.$$

Taking into account the hypercohomology spectral sequence of the functor “sections over U_1' ” with respect to the complex $D F'$:

$$E_2^{pq} = H^p(U_1', \underline{H}^q(D F')) \implies H^*(U_1', D F'),$$

one obtains the relation

$$(***) \quad H^i(U_1', D F') = 0 \quad \text{for } i > -n.$$

Conversely, suppose the preceding relation is satisfied, for every U_1 étale over U , affine over S . Let us apply proposition 3.6, replacing S therein by S' ; the hypotheses of 3.6 concerning $\bar{S}$ are satisfied, since, for every scheme $\bar{U}_1$ étale over $\bar{U}$, affine, one can find a scheme over $\bar{U}_1$ which comes by inverse image from a scheme étale over U , affine over S ; as for the hypotheses concerning F , they are satisfied thanks to 2.3.3. One thus has, for every point u' of U' such that $(\underline{H}^q(D F'))_{\bar{u}'} \neq 0$:

$$\delta_{t'}(u') \leq -n - q,$$

which is nothing other than the relation (**); one has thus proved the equivalence

$$(ii)_t \iff (***).$$

We shall transform the relation (**); one has first

$$H^i(U'_1, D F') = (\underline{H}^i(R f'_{1*}(D F'_1)))_{t'};$$

but by (SGA 5 I 1.12), there exists a canonical isomorphism

$$R f'_{1*}(D F'_1) \simeq D(R f'_1(F'_1)),$$

where $D(R f'_1(F'_1))$ denotes the dual of $R f'_1(F'_1)$ with respect to K . One thus sees that $(ii)_t$ is equivalent to

$$(\underline{H}^i(D(R f'_1(F'_1))))_{t'} = 0 \quad \text{for } i > -n.$$

Applying once more the local duality theorem (SGA 5 I 4.5.3), but this time at the point t' , one finds that

$$(\underline{H}^i(D(R f'_1(F'_1))))_{t'} \simeq D(H_{t'}^{-i}(R f'_1(F'_1))),$$

and finally $(ii)_t$ is equivalent to the relation

$$H_{t'}^i(R f'_1(F'_1)) = 0 \quad \text{for } i < n,$$

i.e. $\text{prof}_{t'}(R f'_1(F'_1)) \geq n$, which completes the proof of the theorem. $\square$

REMARK 4.2.3. The argument simplifies fairly considerably when one assumes that S admits (at least locally) a dualizing complex (for example is locally immersible in a regular scheme). This avoids recourse to a completion (the passage to the case where S is strictly local being immediate), to 2.3.3 and to the rather unappealing technical statement 3.6, which one may then replace by the more congenial reference 3.7.

COROLLARY 4.3. *Let S be an excellent scheme of characteristic zero and $f: U \rightarrow S$ a separated morphism of finite type, such that U is a union of $c + 1$ opens, affine over S . Let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules, with degrees bounded below and with constructible cohomology, n an integer and T a closed subset of S . Suppose that, for every point $u \in U$, one has*

$$\text{prof}_u(F) \geq n - \delta_T(u).$$

Then one has

$$\text{prof}_T(R f(F)) \geq n - c.$$

Indeed, let U_j , $0 \leq j \leq c$, be a covering of U by opens U_j , affine over S . Taking up the notation of the proof of 4.2, one has, for every j ,

$$H^i(U'_j, \underline{H}^q(D F')) = 0 \quad \text{for } i > -n.$$

Using the spectral sequence which relates the cohomology of U to that of the covering formed by the U_j (SGA 4 V 2.4), the preceding relation shows that one has

$$H^i(U', \underline{H}^q(D F')) = 0 \quad \text{for } i > -n + c.$$

The corollary then follows from the end of the proof of 4.2.

COROLLARY 4.4. *Let S be an excellent scheme of characteristic zero, $g: X \rightarrow S$ a morphism, U an open of X , union of $c + 1$ opens affine over S , Y a closed subscheme with underlying space $X - U$ and $j: Y \rightarrow X$ the natural morphism. Let F be a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules on X , with degrees bounded below and with constructible cohomology, T a closed subset of S and n an integer. Suppose that, for every point u of U , one has*

$$\text{prof}_u(F) \geq n - \delta_T(u).$$

Then the canonical morphism

$$H_{g^{-1}(T)!}^i(X/S, F) \longrightarrow H_{(g^{-1}(T) \cap Y)!}^i(Y/S, j^* F)$$

is bijective for $i < n - c - 1$, injective for $i = n - c - 1$.

This follows immediately from 4.1 and 4.3.

COROLLARY 4.5 (Local Lefschetz theorem). *Let S be an excellent henselian local scheme of characteristic zero, t the closed point of S , X a scheme proper over $S' = S - \{t\}$ and U an open of X , union of $c + 1$ affine opens. Let Y be a closed subscheme of X , with underlying space $X - U$, $j: Y \rightarrow X$ the canonical morphism, F a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules on X , with degrees bounded below and with constructible cohomology, and n an integer. Suppose that, for every point u of U , one has*

$$\mathrm{prof}_u(F) \geq n - \delta'_t(u), \quad \text{where } \delta'_t(u) = \delta_t(u) - 1.$$

Then the canonical morphism

$$H^i(X, F) \longrightarrow H^i(Y, j^*F)$$

is bijective for $i < n - c - 1$, injective for $i = n - c - 1$.

Let $f: U \rightarrow S$ be the canonical morphism; it follows from 4.2, applied upon replacing n by $n + 1$, that one has

$$\mathrm{prof}_t(R_! f(F|_U)) \geq n + 1 - c.$$

The preceding relation shows that the canonical morphism

$$H^i(S, R_! f(F|_U)) \longrightarrow H^i(S', R_! f(F|_U))$$

is bijective for $i < n - c$, injective for $i = n - c$. Since $R_! f(F|_U)$ is zero outside S' , one has $H^i(S, R_! f(F|_U)) \simeq (\underline{H}^i(R_! f(F|_U)))_t = 0$, and consequently

$$(*) \quad H^i(S', R_! f(F|_U)) = 0 \quad \text{for } i < n - c.$$

Let $g: X \rightarrow S'$, $h: Y \rightarrow S'$, $f': U \rightarrow S'$ be the canonical morphisms. It follows from the distinguished triangle

$$\begin{array}{ccc} & R h_*(j^*F) & \\ \swarrow & & \searrow \\ R_! f'(F|_U) & \longrightarrow & R g_*(F) \end{array}$$

that the condition $(*)$ is equivalent to the fact that the morphism

$$H^i(S', R g_*(F)) \longrightarrow H^i(S', R h_*(j^*F))$$

is bijective for $i < n - c - 1$, injective for $i = n - c - 1$. Since this morphism identifies canonically with the morphism

$$H^i(X, F) \longrightarrow H^i(Y, j^*F),$$

the conclusion follows at once.

COROLLARY 4.6 (Global Lefschetz theorem). *Let S be the spectrum of a field, X a scheme proper over S and U an open of X union of $c + 1$ affine opens. Let Y be a closed subscheme of X , with underlying space $X - U$, $j: Y \rightarrow X$ the canonical morphism, F a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules on X , with degrees bounded below and with constructible cohomology and n an integer. Suppose that, for every point u of U , one has*

$$\mathrm{prof}_u(F) \geq n - \dim(\overline{\{u\}}).$$

Then the canonical morphism

$$H^i(X, F) \longrightarrow H^i(Y, j^*F)$$

is bijective for $i < n - c - 1$, injective for $i = n - c - 1$.

More generally, if $g: X \rightarrow S$ is a separated morphism of finite type, the hypotheses on S , U , Y , F being the same as above, then the canonical morphism

$$H^i_!(X/S, F) \longrightarrow H^i_!(Y/S, j^*F)$$

(where H_i^i denotes cohomology with proper support, i.e. $H_i^i(X/S, F) = H^i(S, R_i! g(F))$) is bijective for $i < n - c - 1$, injective for $i = n - c - 1$.

The corollary is a special case of 4.4, with $T = S$.

Here is a partial converse to 4.3:

PROPOSITION 4.7. *Let S be a noetherian scheme, $f: U \rightarrow S$ a morphism of finite type. Suppose that there exists a dualizing complex K on S and that $R^! f(K)$ is a dualizing complex on U . Let T be a closed subset of S and c an integer. Then the following conditions are equivalent:*

- (i) *For every complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules F on U , with degrees bounded below and with constructible cohomology, and for every integer n such that one has, for every point u of U ,*

$$\text{prof}_u(F) \geq n - \delta_T(u),$$

one has

$$\text{prof}_T(R^! f(F)) \geq n - c.$$

- (ii) *For every sheaf of $\mathbf{Z}/m\mathbf{Z}$ -modules G on U , constructible, and for every point $t \in T$, one has*

$$(R^p f_*(G))_{\bar{t}} = 0 \quad \text{for } p > \delta(G, f, t) + c$$

(let us recall from (SGA 4 XIX 6.0) that $\delta(G, f, t) = \sup\{\delta_t(u) | t \in \overline{\{u\}} \text{ and } G_{\bar{u}} \neq 0\}$).

N.B. Condition (ii) is satisfied by virtue of 3.1 if f is separated and if U is, locally on S for the étale topology, a union of $c + 1$ opens affine over S , hence 4.7 contains 4.3⁹.

One may evidently assume that S is local and that T is the closed point t of S . The proof of (ii) $\Rightarrow$ (i) is essentially identical to part 2°) of the proof of 4.2. Let us show quickly that (i) $\Rightarrow$ (ii). The local duality theorem (SGA 5 I 4.3.2) applied to $D G$ shows that

$$D(H_u^i(D G)) \simeq (\underline{H}^{-i-2\delta_t(u)}(G))_{\bar{u}}.$$

Since G is reduced to degree 0, one thus has $H_u^i(D G) = 0$ except possibly for $i = -2\delta_t(u)$; more precisely

$$\text{prof}_u(D G) = \begin{cases} -2\delta_t(u) & \text{if } G_{\bar{u}} \neq 0, \\ \infty & \text{if } G_{\bar{u}} = 0. \end{cases}$$

It follows that one has, for any $u \in U$:

$$\text{prof}_u(D G) \geq -n - \delta_t(u).$$

It then follows from hypothesis (i) that one has $\text{prof}_t(R^! f(D G)) \geq -n - c$. One transforms this relation using the isomorphism $R^! f(D G) \simeq D(R f_*(G))$ (SGA 5 I 1.12) and applying the local duality theorem at the point t ; one thus obtains

$$(\underline{H}^i(R f_*(G))_{\bar{t}}) = 0 \quad \text{for } i > n + c,$$

which is nothing other than (ii).

4.8. The hypotheses being those of 4.4 with g proper (resp. 4.5, resp. 4.6 with g proper), if V is an open neighbourhood of Y in X , the morphism

$$H^i(V, F) \longrightarrow H^i(Y, j^* F)$$

is bijective for $i < n - c - 1$, injective for $i = n - c - 1$. If $\iota: V \rightarrow X$ is the canonical morphism, it suffices indeed, in order to see this, to apply 4.4 (resp. 4.5, resp. 4.6) to the complex $R \iota_*(F|_V)$. One may ask whether the preceding morphism is bijective if $i = n - c - 1$, injective for $i = n - c$. It is evidently sufficient that the hypotheses be satisfied when one replaces n by $n + 1$; the following proposition shows that a little less suffices.

⁹At least in the case where S locally admits a dualizing complex, for example S locally immersible in a regular scheme.

PROPOSITION 4.9. *Let S be an excellent local scheme of characteristic zero, with closed point t (resp. in addition to the preceding conditions one assumes S henselian), $f: X \rightarrow S$ a scheme proper over S (resp. proper over $S - \{t\}$) and U an open of X union of $c + 1$ affine opens. Let Y be a closed subscheme of X , with underlying space $X - U$, $j: Y \rightarrow X$ the canonical morphism, F a complex of sheaves of $\mathbf{Z}/m\mathbf{Z}$ -modules on X , with degrees bounded below and with constructible cohomology, and n an integer. One assumes that one has, for every point u of U ,*

$$\text{prof}_u(F) \geq \inf(n - 1, n - \delta_t(u)) \quad (\text{resp. } \text{prof}_u(F) \geq \inf(n - 1, n + 1 - \delta_t(u))).$$

Then for every open neighbourhood V of Y in X , the canonical morphism

$$H_{f^{-1}(t)}^i(V, F) \longrightarrow H_{f^{-1}(t) \cap Y}^i(Y, j^*F) \quad (\text{resp. } H^i(V, F) \longrightarrow H^i(Y, j^*F))$$

is bijective for $i < n - c - 2$ and injective for $i = n - c - 2$. Moreover, there exists an open neighbourhood V_0 of Y in X such that, for every other such V with $V \subset V_0$, the canonical morphism

$$H_{f^{-1}(t) \cap V}^i(V, F) \longrightarrow H_{f^{-1}(t) \cap Y}^i(Y, j^*F) \quad (\text{resp. } H^i(V, F) \longrightarrow H^i(Y, j^*F))$$

is bijective for $i < n - c - 1$, injective for $i = n - c - 1$.

PROOF. To simplify, set $\delta'_t(u) = \delta_t(u)$ (resp. $\delta'_t(u) = \delta_t(u) - 1$). One deduces from 4.8 the first assertion of 4.9, since the hypotheses of 4.4 (resp. 4.5) are satisfied when one replaces n therein by $n - 1$. They are also satisfied for n itself, except at the points u such that $\delta'_t(u) = 0$. Now, for a $u \in U$, to say that one has $\delta'_t(u) = 0$ is equivalent to saying that u is a *closed point of U_t* (resp. a *closed point of X*). Let E be the set of points of U such that $\delta'_t(u) = 0$; let us show that, *for all points $u \in E$, except a finite number, one has $\text{prof}_u(F) \geq n$* . Let $\bar{S}$ be the strict localization of S at t , S' the completion of $\bar{S}$, with closed point t' , and consider the cartesian square

$$\begin{array}{ccc} U' & \xrightarrow{h} & U \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S. \end{array}$$

The depth hypotheses at the points of U are preserved when one replaces U by U' and F by the inverse image F' of F on U' . Indeed, let $u' \in U'$ and $u = h(u')$. If $u \notin E$, one has the relation $\text{prof}_u(F) \geq n - \delta'_t(u)$, and it follows from 4.2.1 that this implies the relation $\text{prof}_{u'}(F') \geq n - \delta'_t(u')$. If $u \in E$, u' is a closed point of U'_t (resp. a closed point of $X' = X \times_S S'$), and, since the fibre U'_u of h at u is of dimension zero and h is regular, it follows from 1.16 that one has $\text{prof}_{u'}(F') = \text{prof}_u(F) \geq n - 1$.

Let then K be a dualizing complex on S' , normalized at the closed point t' , and $D F'$ the dual of F' with respect to $R^1 f(K)$. According to 4.2.2, the étale depth hypotheses at the points of U' translate into the relations:

$$(\underline{H}^q(D F'))_{\bar{u}'} = 0 \quad \text{for } q > -n - \delta'_t(u') \quad (\text{resp. } q > -n - 2 - \delta'_t(u')),$$

if u' is not a point of $E' = h^{-1}(E)$,

$$(\underline{H}^q(D F'))_{\bar{u}'} = 0 \quad \text{for } q > -(n - 1) \quad (\text{resp. } q > -n - 1), \text{ if } u' \in E'.$$

Let $G = \underline{H}^{-(n-1)}(D F')$ (resp. $G = \underline{H}^{-n-1}(D F')$); since G is a constructible sheaf, the set of points at which the geometric fibre is nonzero is a constructible set (SGA 4 IX 2.4 (iv)); now by hypothesis this set is contained in the set E' of closed points of U'_t (resp. of points of U' closed in X'); it therefore follows from 4.9.1 below that this set is reduced to a finite number of points. Applying 4.2.2, one sees that, for all points of E' , except a finite number, one has $\text{prof}_{u'}(F') \geq n$. It follows indeed by 1.16 that, for all points of E except a finite number, one has

$$\text{prof}_u(F) \geq n.$$

Let V be an open neighbourhood of Y in X , contained in the complement in X of the finite set of points u of E for which one has $\text{prof}_u(F) = n - 1$. If $\iota: V \rightarrow X$ is the canonical immersion, let

$$F_1 = R\iota_*(F|_V) ;$$

then F_1 is a complex of sheaves on X , with constructible cohomology (SGA 4 XIX 5.1) and with degrees bounded below. We are going to see that, for every point u of U , the complex F_1 satisfies the relation

$$(*) \quad \text{prof}_u(F_1) \geq n - \delta'_t(u).$$

If $u \in U \cap V$, one has $\text{prof}_u(F_1) = \text{prof}_u(F)$, and the relation $(*)$ is satisfied by hypothesis on the points of U which do not belong to E ; for the latter, it is also satisfied according to the choice of V . Finally, if $u \in U$ and $u \notin V$, one has according to 1.6 g) $\text{prof}_u(F_1) = \infty$. One then applies 4.4 (resp. 4.5) upon replacing F by F_1 ; one obtains the stated result, taking into account the fact that one has, for any i :

$$H_{f^{-1}(t)}^i(X, R\iota_*(F|_V)) \simeq H_{f^{-1}(t) \cap V}^i(V, F) \quad (\text{resp. } H^i(X, R\iota_*(F|_V)) \simeq H^i(V, F)).$$

□

LEMMA 4.9.1. *A constructible set E contained in the set of closed points of a noetherian scheme X is reduced to a finite number of points.*

Indeed, E is a finite union of sets of the form $U \cap \mathbb{C}V$, where U and V are opens of X ; by hypothesis all the points of $U \cap \mathbb{C}V$ are maximal points of this set, hence they are finite in number.

5. Geometric depth

To apply 4.2 and its corollaries in practice, one needs a convenient criterion allowing one to verify the étale depth hypotheses at the points of U . We shall give such a criterion, using the local Lefschetz theorem 4.5.

5.1. Let A be a noetherian local ring; when we speak of the étale depth of A , it will be the depth at the closed point. We shall introduce a notion of “geometric depth of A ”, and use 4.5 to compare it with the étale depth $\text{et. prof}(A)$.

PROPOSITION 5.2. *Let A be a noetherian local ring; suppose that A is isomorphic to a quotient of a regular local ring B by an ideal I (this is true for example when A is complete, by Cohen’s theorem (EGA 0_{IV} 19.8.8)). Let q be the minimal number of generators of I ; then the number $\dim(B) - q$ is independent of the choice of B .*

The minimal number of generators of I is also equal to the rank of the k -vector space $I \otimes_B k$, where k denotes the residue field of A . One reduces immediately to the case where A is complete, since one has $\widehat{A} \simeq \widehat{B}/\widehat{I}$ with $\dim \widehat{B} = \dim B$ and $\text{rg}_k(I \otimes_B k) = \text{rg}_k(\widehat{I} \otimes_{\widehat{B}} k)$; for the same reason one may suppose that the ring B is complete. Let B and B' be two complete regular local rings, $f: B \rightarrow A$, $f': B' \rightarrow A$ two surjective homomorphisms and $I = \text{Ker}(f)$, $I' = \text{Ker}(f')$. One must show that

$$\dim B - \text{rg}_k(I \otimes_B k) = \dim B' - \text{rg}_k(I' \otimes_{B'} k).$$

Let us first consider the case where one has a factorization of the form

$$\begin{array}{ccc} B & \xrightarrow{f} & A \\ & \searrow g & \nearrow f' \\ & B' & \end{array}$$

with g surjective. Let $J = \text{Ker}(g)$; then $J \subset I$ and $I/J = I'$. Since B' is regular, $\dim(B') = \dim(B) - \text{rg}_k(J \otimes_B k)$ and J is generated by elements forming part of a regular system of parameters of B . It follows that one has the exact sequence

$$0 \longrightarrow J \otimes_B k \longrightarrow I \otimes_B k \longrightarrow J/I \otimes_{B'} k \longrightarrow 0,$$

and consequently

$$\dim B - \text{rg}_k(I \otimes_B k) = \dim B - \text{rg}_k(J \otimes_B k) - \text{rg}_k(J/I \otimes_{B'} k) = \dim B' - \text{rg}_k(I' \otimes_{B'} k).$$

The general case reduces to the preceding one; to see this, it suffices to show that one can find a complete regular local ring B'' and surjective homomorphisms $g: B'' \rightarrow B$ and $g': B'' \rightarrow B'$, making the diagram commutative

$$(*) \quad \begin{array}{ccc} & B & \\ g \nearrow & & \searrow f \\ B'' & & A \\ g' \searrow & & \nearrow f' \\ & B' & \end{array}$$

Now, if W is a Cohen ring with residue field k , one has a local morphism $W \rightarrow A$ which lifts to B and B' (EGA IV 19.8.6), so that one has the commutative diagram

$$\begin{array}{ccc} & B & \\ W \nearrow & & \searrow \\ & A & \\ W \searrow & & \nearrow \\ & B' & \end{array}$$

One can find integers n and n' and surjective morphisms $h: W[[T_1, \dots, T_n]] \rightarrow B$ and $h': W[[T'_1, \dots, T'_{n'}]] \rightarrow B'$ which are morphisms of W -algebras (EGA 0_{IV} 19.8.8); if one then sets $B'' = W[[T_1, \dots, T_n, T'_1, \dots, T'_{n'}]]$ and if one defines g and g' as morphisms of W -algebras such that

$$g(T_i) = h(T_i), \quad g(T'_i) = b_i, \quad g'(T_i) = b'_i, \quad g'(T'_i) = h'(T'_i),$$

where b_i (resp. b'_i) is an element of B (resp. of B') which lifts $(f' \circ h')(T'_i)$ (resp. $(f \circ h)(T_i)$), the diagram $(*)$ is indeed commutative.

Proposition 5.2 justifies the following definition:

DEFINITION 5.3. Let A be a noetherian local ring, $\hat{A}$ its completion, which is thus isomorphic to the quotient of a complete regular local ring B by an ideal I ; if q is the minimal number of generators of I , one calls the *geometric depth* of A the number

$$\text{geom. prof}(A) = \dim B - q.$$

PROPOSITION 5.4. *Let A be a noetherian local ring. Then one has*

$$\text{geom. prof}(A) \leq \dim A,$$

and one has equality if and only if A is a complete intersection.

One may suppose A complete. Let then $A = B/I$, where B is a complete regular local ring and I an ideal of B . If $(x_1, \dots, x_q)$ is a minimal system of generators of I , one has $\dim A \geq \dim B - q$, and to say that $\dim A = \dim B - q$ is equivalent to saying that $(x_1, \dots, x_q)$ forms part of a system of parameters of B (EGA 0_{IV} 16.3.7); the proposition follows immediately.

PROPOSITION 5.5. *Let A and A' be noetherian local rings, $f: A \rightarrow A'$ a local homomorphism. Suppose that f is flat and that, denoting by k the residue field of A , $A' \otimes_A k$ is a field, a separable extension of k . Then one has*

$$\text{geom. prof}(A) = \text{geom. prof}(A').$$

Replacing A and A' by their completions if necessary, one may suppose A and A' complete (it follows from (EGA 0_{III} 10.2.1) that the flatness hypothesis is preserved and this is evident for the other hypotheses). Let then $A = B/I$, where B is a regular local ring and I an ideal of B . Since A' is formally smooth over A (EGA 0_{IV} 19.8.2), it follows from (EGA 0_{IV} 19.7.2) that one can find a complete noetherian local ring B' and a local homomorphism $B \rightarrow B'$, such that B' is a flat B -module and that one has $B' \otimes_B A \simeq A'$. One thus has $A' \simeq B'/IB'$; moreover the ring B' is regular; indeed, if $\mathfrak{m}$ is the maximal ideal of B , $\mathfrak{m}B'$ is the maximal ideal of B' , and, since $\mathfrak{m}$ is generated by a regular sequence by definition of “regular”, $\mathfrak{m}B'$ is generated by a B' -regular sequence (EGA 0_{IV} 15.1.14). Since one evidently has $\dim B = \dim B'$, and since I and IB' have the same minimal number of generators, the assertion follows.

THEOREM 5.6¹⁰. *Let A be an excellent local ring of characteristic zero. Then one has*

$$\text{et. prof}(A) \geq \text{geom. prof}(A).$$

One may suppose A strictly local complete, since geometric depth and étale depth are preserved by passage to the strict henselization and to the completion by 5.5 and 1.16. Let $A \simeq B/I$, where B is a complete regular local ring, and let $(f_1, \dots, f_q)$ be a minimal system of generators of the ideal I . One thus has

$$\pi = \text{geom. prof}(A) = \dim B - q.$$

Consider the closed immersion

$$Y = \text{Spec } A \longrightarrow X = \text{Spec } B,$$

and let $U = X - Y = \bigcup_{1 \leq i \leq q} X_{f_i}$. If a denotes the closed point of X , one must show that, for every prime number p , one has

$$H_a^i(Y, \mathbf{Z}/p\mathbf{Z}) = 0 \quad \text{for } i < \pi.$$

Since B is regular excellent, one has $\text{et. prof}(B) = 2 \dim X$ (cf. 1.10) and consequently $H_a^i(X, \mathbf{Z}/p\mathbf{Z}) = 0$ for $i < 2 \dim X$. It therefore suffices, to prove the theorem, to prove that the morphism

$$(*) \quad H_a^i(X, \mathbf{Z}/p\mathbf{Z}) \longrightarrow H_a^i(Y, \mathbf{Z}/p\mathbf{Z})$$

is bijective for $i < \pi$. For this one applies the local Lefschetz theorem 4.5 with $n = \pi + q - 1$, $c = q - 1$ hence $n - c = \pi$. Note that $U = X - Y$ is the union of the q affine open sets X_{f_i} . Let us show that one has, for every point u of U :

$$\text{et. prof}_u(X) \geq \pi + q - 1 - \dim(\overline{\{u\}}) = \dim \mathcal{O}_{X,u}$$

(where $\overline{\{u\}}$ denotes the closure of u in $X - \overline{\{a\}}$). Indeed it follows from 1.10 that one has

$$\text{et. prof}_u(X) = 2 \dim \mathcal{O}_{X,u} \geq \dim \mathcal{O}_{X,u}.$$

Using 4.5, one sees that $(*)$ is bijective for $i < \pi$, which completes the proof of the theorem.

COROLLARY 5.7. *Let S be the spectrum of a field of characteristic zero (resp. a henselian excellent local scheme of characteristic zero), $f: X \rightarrow S$ a scheme proper over S (resp. over $S - \{s\}$). Let U be a union of $c + 1$ open sets of X , affine, Y a closed subscheme with underlying space $X - U$, n and m integers > 0 . Suppose that, for every point u of U , one has*

$$\text{geom. prof}(\mathcal{O}_{X,u}) \geq n - \dim(\overline{\{u\}})$$

($\overline{\{u\}}$ the closure of u in X). Then the canonical morphism

$$H^i(X, \mathbf{Z}/m\mathbf{Z}) \longrightarrow H^i(Y, \mathbf{Z}/m\mathbf{Z})$$

is bijective for $i < n - c - 1$, injective for $i = n - c - 1$.

¹⁰**N.D.E.** Illusie has since shown the inequality $\text{prof}_x(\mathbf{Z}/\ell^\nu\mathbf{Z}) \geq \text{geom. prof}_x(X/S) - \delta(x) + 1$ for x a point of X a scheme of finite type over a trait S of residual characteristic prime to ℓ , and $\nu \geq 1$. If S is of characteristic zero, this is a consequence of theorem 5.6; see (Illusie L., “Perversité et variation”, *Manuscripta Math.* **112** (2003), p. 271–295).

One applies 4.5 and 4.6. The étale depth hypotheses at the points of U are satisfied since one has by 5.6

$$\text{et. prof}_u(X) \geq \text{geom. prof}(\mathcal{O}_{X,u}) \geq n - \dim(\overline{\{u\}}).$$

6. Open questions

6.1. One may ask whether the implication (ii) $\Rightarrow$ (i) of 4.2 is valid more generally for torsion sheaves F , not necessarily annihilated by a given integer m and not necessarily constructible. In the case where S is not of characteristic zero, it seems possible that this implication remains valid, even for sheaves of p -torsion (p the residual characteristic). Finally it is also not clear that the hypothesis that S be excellent cannot be removed.

6.2. Let X be a scheme proper over a field k or else the complement of the closed point of a henselian local scheme, and $j : Y \rightarrow X$ a closed subscheme of X , whose complement U is affine. Then, if F is a sheaf of sets on X or a sheaf of not necessarily commutative groups, the statements 4.5 or 4.6 and 4.9 still make sense for such an F , provided one restricts oneself to small values of n . If u is a point of U , one denotes by $\bar{u}$ a geometric point over u , by $X(\bar{u})$ the strict localization of X at $\bar{u}$ and by $F_{\bar{u}}$ the fibre of F at $\bar{u}$. Then, possibly making certain hypotheses on X and on F , for example assuming X excellent (possibly of characteristic zero, or of equal characteristic using resolution of singularities) and F ind-finite (or if need be even $\mathbf{L}$ -ind-finite with $\mathbf{L}$ prime to the characteristic of X), one would like to prove the following statements:

- a) Let F be a sheaf of sets (resp. a sheaf of groups) and suppose that, for every point u of U , one has

$$F_{\bar{u}} \longrightarrow H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) \leq 1$$

(that is, for such a u , one has $\text{prof}_u(F) \geq 1$). Then, as V runs through the set of open neighbourhoods of Y , the canonical morphism

$$\varinjlim_V H^0(V, F) \longrightarrow H^0(Y, j^* F)$$

is bijective (resp. one has the preceding conclusion and moreover the morphism $\varinjlim_V H^1(V, F) \rightarrow H^1(Y, j^* F)$ is injective). If F is constructible, one may replace the $\varinjlim$ by the cohomology of V for V “small enough”.

- b) Let F be a sheaf of sets (resp. a sheaf of groups) and suppose that, for every point u of U , one has $\text{prof}_u(F) \geq 2 - \dim(\overline{\{u\}})$, which is also expressed by the relations

$$F_{\bar{u}} \longrightarrow H^0(X(\bar{u}) - \bar{u}, F) \text{ is bijective if } \dim(\overline{\{u\}}) = 0$$

$$F_{\bar{u}} \longrightarrow H^0(X(\bar{u}) - \bar{u}, F) \text{ is injective if } \dim(\overline{\{u\}}) = 1.$$

Then the canonical morphism

$$H^0(X, F) \longrightarrow H^0(Y, j^* F)$$

is bijective (resp. one has the preceding conclusion and moreover the morphism $H^1(X, F) \rightarrow H^1(Y, j^* F)$ is injective).

- c) Let F be an ind-finite sheaf of groups. Suppose that, for every point u of U , one has

$$F_{\bar{u}} \longrightarrow H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective if } \dim(\overline{\{u\}}) = 0 \text{ or } 1,$$

$$F_{\bar{u}} \longrightarrow H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) = 2.$$

Then, as V runs through the set of open neighbourhoods of Y , the canonical morphisms

$$\varinjlim_V H^0(V, F) \longrightarrow H^0(Y, j^* F) \text{ and } \varinjlim_V H^1(V, F) \longrightarrow H^1(Y, j^* F)$$

are bijective. If F is constructible, one may replace the $\varinjlim$ by the cohomology of V for V small enough.

- d) Let F be a sheaf of groups. Suppose that, for every point u of U , one has $\text{prof}_u(F) \geq 3 - \dim(\overline{\{u\}})$, which is also expressed by the conditions

$$F_{\bar{u}} \longrightarrow H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective, and } H^1(X(\bar{u}) - \bar{u}, F) = 0 \text{ if } \dim(\overline{\{u\}}) = 0,$$

$$F_{\bar{u}} \longrightarrow H^0(X(\bar{u}) - \bar{u}, F) \text{ bijective if } \dim(\overline{\{u\}}) = 1,$$

$$F_{\bar{u}} \longrightarrow H^0(X(\bar{u}) - \bar{u}, F) \text{ injective if } \dim(\overline{\{u\}}) = 2.$$

Then the canonical morphisms

$$H^0(X, F) \longrightarrow H^0(Y, j^*F) \text{ and } H^1(X, F) \longrightarrow H^1(Y, j^*F)$$

are bijective.

As an indication in favour of these statements¹¹, we mention ?? ??, ?? ?? and ?? ??. We mention that, thanks to the argument of 4.8 and 4.9, statement a) (resp. c)) would follow from b) (resp. d)).

6.3. There would follow from d) the following statement, analogous to 5.6: if A is a noetherian local ring (possibly excellent) and if one has $\text{prof geom}(A) \geq 3$, then one has $\text{prof hop}(A) \geq 3$. To see this, one realises $Y' = \text{Spec } A$ as a closed subscheme of a regular local scheme $X' = \text{Spec } B$, whose complement is a union of q affine open sets, with the relation $\dim B - q = \text{prof geom}(A)$. One has, for every point x of X' , if $n = \dim B$, $\text{prof hop}_x(X) \geq \inf(3, n - \dim(\overline{\{x\}}))$ (cf. 1.11) and one deduces from d) that this implies $\text{prof hop}_y(Y') \geq \inf(3, n - q - \dim(\overline{\{y\}}))$, for every point y of Y' . The result is then obtained by taking for y the closed point of Y' .

6.4. A variant of 4.2, at least of the implication (ii) $\Rightarrow$ (i), should still be valid in the complex analytic case, provided one works with “analytically constructible” sheaves (cf. ??); the proof would be analogous to that of 4.2, using the duality theory of J.-L. Verdier. Note on the other hand that, for the complex analytic analogue of the non-commutative variants indicated in 6.2, one does not even have a method of attack for the statements concerning the fundamental group suggested by the results of Exposés ??, ??, ??, recalled at the end of 6.2. The methods of the Seminar seem in fact to be irremediably tied to the case of *finite* coverings (which can be studied in terms of coherent sheaves of algebras).

¹¹**N.D.E.** all the statements of 6.2, apart from the constructible variants, have been proved by Mme Raynaud; see (Raynaud M., “Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental”, *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52, corollary III.1.3).

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